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SoilSense: Machine Learning-Based Soil Health Classification and Crop Fertilizer Recommendation System

Capstone Project

Presented to the Faculty of the State University

Polytechnic University of the Philippines

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ABSTRACT

An important factor in agricultural productivity and sustainability is the health of the soil. Unfortunately, many small-scale farmer communities do not have easy access to a reliable soil analysis facility that provides affordable advice on the use of fertilizers. Conventional analysis of the soil properties takes a long time and also results in the misuse of fertilizers as well as a decrease in crop production. The capstone project proposes a decision support system called “SoilSense” that uses the “IoT” technology and machine learning algorithm to classify the health of the soil in real time through an Android application.

This system relies on soil sensors to determine the important parameters like pH value, moisture, and temperatures, as well as the values for nutrients like nitrogen, phosphorus, and potassium. This information is fed to a Firebase platform via an ESP32 microcontroller and is analyzed by a TensorFlow-based machine learning algorithm to determine the soil status. According to the classification, fertilizer recommendations for the specific plants are made by the system based on the data obtained. This mobile application also enables the user to check the data and history related to the soils.

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The goal of SoilSense is to facilitate precision agriculture through the efficient use of resources, increased crop yield, and sustainable practices, especially in the case of farmers in remote areas such as Barangay Rizal in Lopez, Quezon. Despite its restrictions in anything related to the precision of the sensors, internet connections, or environmental influences that are beyond its control, it has shown that combining IoT technology and machine learning can be an effective way to deal with the pressing issues concerning agriculture in the modern world.

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CHAPTER 1 – INTRODUCTION

1.1 PROJECT CONTEXT

Agriculture is still the pillar of rural livelihoods and food security. Soil degradation, nutrient loss, and improper use of fertilizer have contributed to decreasing soil health and declining crop productivity in most areas. Conventional methods for assessing soil quality, such as laboratory testing by hand, are often labor-intensive, expensive, and not accessible for smallholder farmers. Recent advancements in machine learning (ML) offer the chance to automate and improve soil classification based on measurable soil properties like pH, nitrogen, phosphorus, potassium, and humidity levels. With the training of smart models on past soil and crop information, it is now possible to develop decision-making instruments that not only classify the level of soil health but also suggest suitable fertilizer varieties and crop choices with respect to the condition of the soil. This project attempts to respond to this critical need through an intelligent, scalable system that will contribute to sustainable agriculture by tapping into machine learning for real-time, data-informed insights related to soil health and fertilization.

This project aims to meet the urgent need for a smart, scalable system that supports sustainable agriculture. It uses machine learning to offer data-driven, real-time information about soil conditions and fertilization. It will also help in the standardization of local farming practices, considering local crop profiles relevant to local areas, such as Barangay Rizal, where different varieties of Talong are usually planted.

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Pallavi et al. 2024 presents a machine learning-based system. It recommends optimal crops and fertilizers considering major soil characteristics and weather conditions. This study aims to highlight the role of precision agriculture in encouraging sustainable farming methods. Using several ML models, the system generates site-specific suggestions that are intended to improve agricultural yield while maintaining soil health. . It takes into consideration environmental factors like soil pH, temperature, humidity, and nutrient levels for selecting the best crop-fertilizer matching.

The project develops the science of sustainable agriculture through applying IoT monitoring systems to overcome the technical and practical challenges of culture by introducing

IoT monitoring systems to overcome the technical and practical challenges of traditional SoilSense. It also presents a model for future development that promotes conservation of resources and environmental responsibility.

1.2 TECHNICAL BACKGROUND

The proponents had gathered all the related research about the project technicality and had brainstormed about the technologies that will be used in the capstone project. The documentation of the proponents of Arduino parts hardware and software that can help the improvement of the development of the SoilSense systems.



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1.2.1 HARDWARE/ EQUIPMENT

Microcontroller

The microcontroller is the core of the system, **ESP32 DevKit** can be regarded as the main controller that fulfills all the functions. But this controller has a development board that includes many characteristics that are suitable for advanced projects that require higher processing power and connectivity

Soil Comprehensive Sensor

The sensor that the researchers use is **Soil Comprehensive Sensor**, it is a multi-parameter sensing device that detects and transmits key soil properties such as **humidity, temperature, pH, and nutrient levels (NPK)**. The given sensor provides real-time data in an application-based system to the microcontroller ESP32, which then processes the information and relays it to a mobile application. The data enables users to monitor soil health for classification of stages of the crop and fertilizer applications.

User Device

A User Device refers to any interface device that allows the end user to access, monitor, or interact with data collected by the microcontroller system. This can be a smartphone or tablet that connects via Wi-Fi, or the internet that will be used to display data via a mobile app.



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1.2.2 SOFTWARE

Arduino IDE

This will be the Integrated Development Environment (IDE) the researchers will use for writing the program for the microcontroller of the Arduino Board.

Android Studio

This is the Integrated Development Environment (IDE) the researchers will use to create the mobile application. Through this environment, the researchers use Kotlin and Java, which are the main source codes for developing this project's app.

Cloud Backend

The backend that researchers will be using is a Firebase to store required information, which can then be used for showing real-time sensor data, and integrating its other services like Firebase Real Time Database. It allows them to manage extensive amounts of information, thus leading to swift progress during establishment.

Kotlin and Java

The programming languages involved in building a mobile application are these two, because they are best suited to Android Studio. They also have several useful features that make it possible to complete an app much more quickly.



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1.2.3 PEOPLEWARE/MANPOWER

Capstone project "SoilSense" and the product's continuous use require a multidisciplinary team. The Software Developer must play the role of the designer, programmer, and tester of the software aspects of the IoT monitoring system and of the user interface. The hardware Developer's job is particularly aimed at selecting the components, coupling and interfacing them with the hardware devices such as sensors, micro controllers, total structure and design of the product. But the Writer is the individual that deals with the Documents throughout the research and development process, such as methodology, results, limitations, and possible improvements. This job is crucial in preparing technical reports, presentations, and academic publications.

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1.2.4 NETWORK INFRASTRUCTURE/ARCHITECTURE

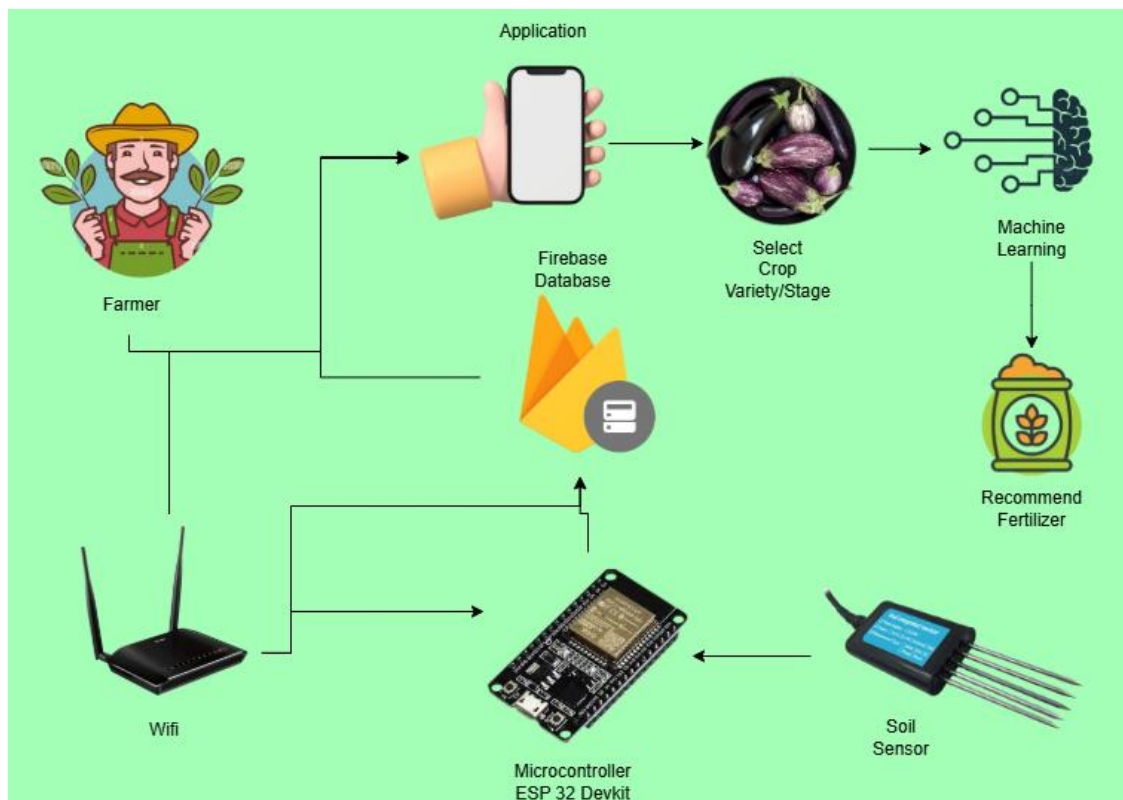


Figure 1.2.4-1 Network Infrastructure Diagram

On the network infrastructure and architecture side, the researchers have taken the approach of a system initially with soil sensors that identify the key parameters of NPK, pH, and moisture content. ESP32 microcontrollers are used to take readings from these sensors due to their flexibility and cross-platform compatibility in IoT use cases. The ESP32 acts as the central controller to control communication between sensors and the internet. The gathered data is then sent to the cloud storage server over the internet through Firebase. Firebase is a scalable, real-time platform that serves as a central hub from which data is stored and communicated. The app then fetches the readings to have the machine learning model analyze



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the received data of the soil for the most appropriate fertilizer for the soil and how much fertilizer is needed.

After the analysis is finished, the system automatically delivers the suggestions which are then displayed to the user's application, facilitating timely and well-informed agricultural decisions.

1.2.5 STORAGE, BACKUP AND RECOVERY PROCEDURE

The researchers have selected Firebase as the major platform for the Storage, Backup and Recovery Procedure of data gathered by the system. Due to its real-time features and integrated data management capabilities, it is a suitable solution for handling the constant flow of information from the soil sensors and the IoT infrastructure.

Firebase Cloud plays a key role in enabling smooth integration between the smart agriculture recommendation platform and the soil monitoring system. Through Firebase, the data collected by ESP32 microcontrollers from the soil sensors, such as NPK contents, pH levels, and water content, is shared and securely stored in the cloud. Machine learning algorithms running in the cloud then use this data to provide accurate recommendations for suitable crops and fertilizers based on the identified soil properties. In addition, Firebase offers native data export support, automatic backup, and recovery features. They protect data from loss and ensure seamless system operation even in the occurrence of connectivity or hardware failure. The scalability



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and stability of Firebase infrastructure guarantee that all such essential data from raw sensor inputs to AI suggested recommendations are secure, accessible, and recoverable at any moment.

Its high availability, cross-platform portability, and easy integration with mobile apps make Firebase a good foundation for this IoT-based smart farming system's data processing requirements.

1.2.6 SECURITY PROCEDURES

For secure system operation, the researchers have put in place fundamental protective mechanisms in both hardware and software aspects.

Although the mobile app does not support user authentication, sensor data and communication integrity are still protected. Data transmitted from the ESP32 microcontroller to the cloud is encrypted, while also avoiding unauthorized interception, to ensure that agricultural sensor data remains secure and accurate during transmission.

Internet access is provided to the ESP32 microcontrollers, which are the special communication gateway between soil sensors and the cloud network. This approach limits excessive exposure of other devices to the internet, limiting possible vulnerabilities and external attacks.

By limiting data transmission to secure and controlled connectivity, the system preserves robustness, providing guaranteed performance for real-time agricultural



monitoring and recommendation activities without the need for complicated login processes for end users.

1.2.7 POLICIES AND PROCEDURES

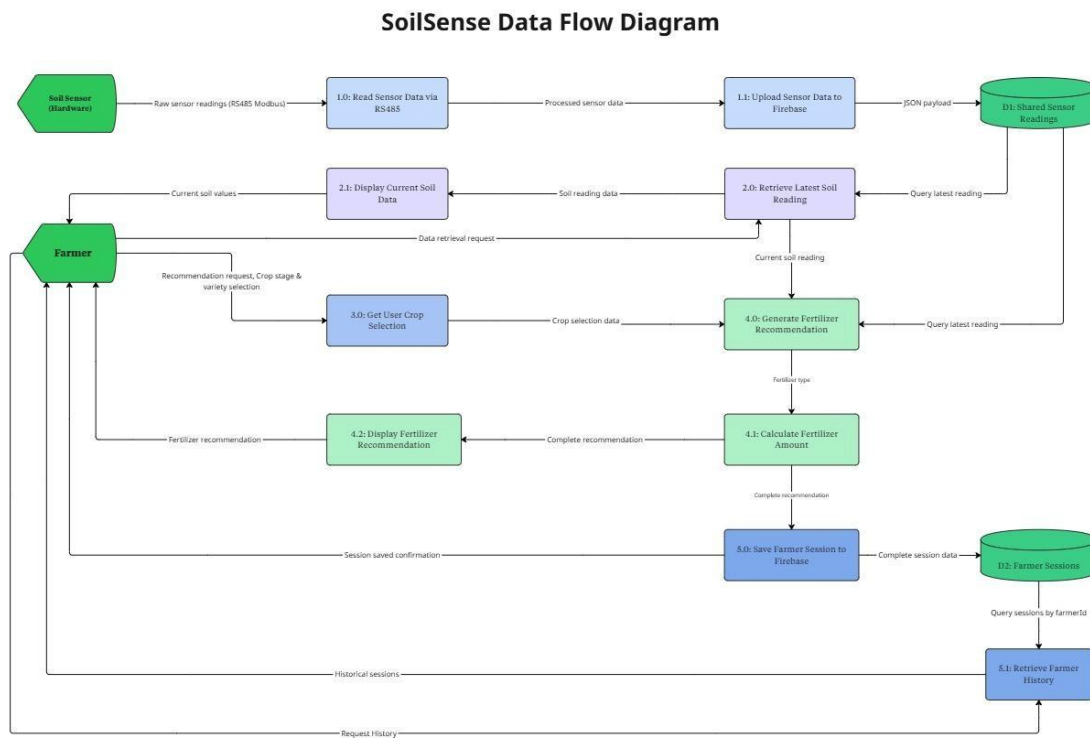


Figure 1.2.7-1 Data Flow Diagram

The Data Flow Diagram of the SoilSense System shows how information travels through the different parts of the platform. It starts with collecting soil data and ends with generating and storing fertilizer recommendations.

The process begins at the Soil Sensor, which measures real-time soil parameters like moisture, pH, temperature, and nutrient levels. These raw readings are sent to the



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ESP32 microcontroller, which receives and processes the data. After this, the ESP32 uploads the sensor readings to the Firebase Realtime Database, where they are stored as structured data.

Next, the SoilSense Mobile Application retrieves the latest soil values from Firebase. This data is sent to the system's TensorFlow-based Machine Learning Model, which examines the soil condition and decides on the best fertilizer type for the user's selected crop. The application then shows the generated fertilizer recommendation, including the type of fertilizer and the estimated amount.

Once the recommendation is viewed, the system saves the entire session, which includes soil readings, crop selection, and ML-generated advice, back to Firebase. This allows farmers to access their past recommendations at any time through the app. This DFD shows a continuous flow of data: sensor readings are captured, sent, analyzed, and stored in an organized way. This ensures that farmers receive data-driven fertilizer advice while keeping a record of their past soil evaluations and recommendations.

1.2.8 DATA AND PROCESS

To design the SoilSense, begin by declaring the purpose of the system. The IoT sensor in this case identifies the data by gathering different parameters from the soil like humidity content, pH level, temperature and nutrient levels. However, the process is outlined by the collection of data afterwards to process it and save data to the database. Whereas the ML classification is to recommend the crop, fertilizer and



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irrigation guidance based on the soil which machine learning classifies. Subsequently, the data is directly informed to the user's device, where they can track the present status.

1.3 PROBLEM ANALYSIS

Fishbone Diagram

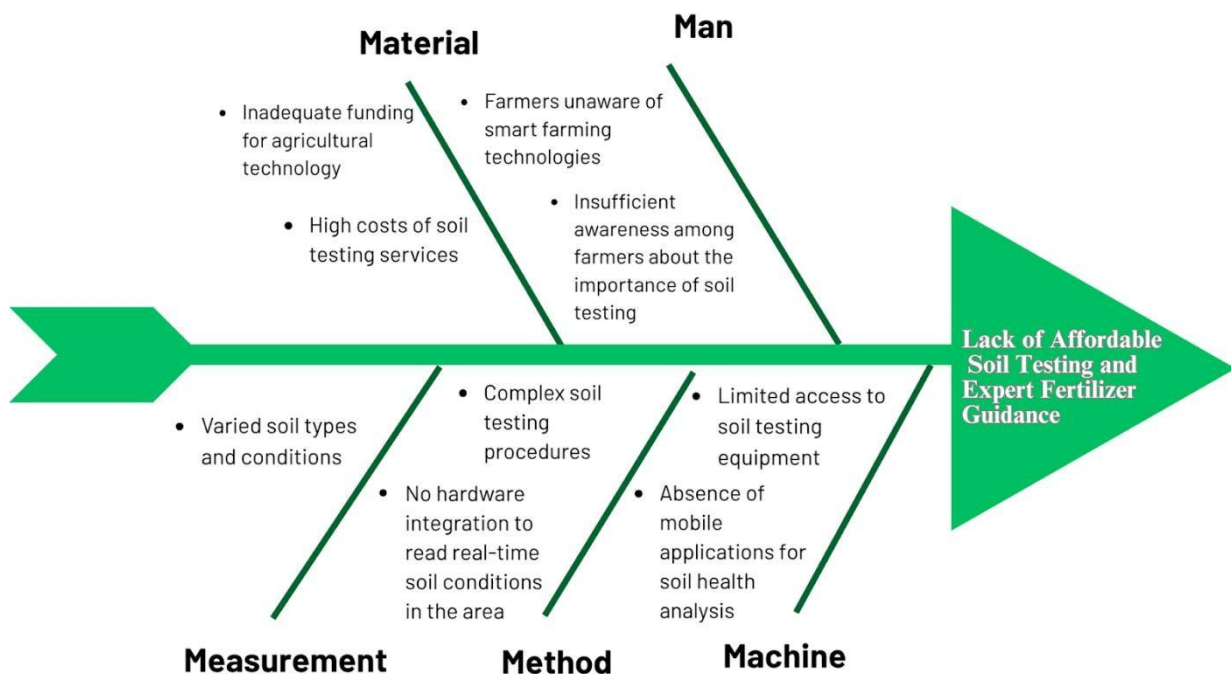


Figure 1.3.1-1. Fishbone Diagram of SoilSense

The fishbone diagram shows the underlying reasons for the unaffordable soil testing and professional fertilizer advice by breaking them down into five broad categories: Material, Man, Measurement, Method, and Machine. Under the category of Material are the insufficient funds allocated for agro-technology and high charges for soil testing services, which make the resources unaffordable for most farmers. The



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Man category indicates the shortage of skilled agricultural professionals and inadequate knowledge among farmers regarding the significance of soil testing. Challenges of a measurement nature include heterogeneous soil conditions and types, as well as climatic conditions influencing soil health, making the process of testing them complex. Challenges of a method nature involve intricate procedures in soil testing and ineffective channels of communication among farmers and experts, which interfere with timely information sharing. Finally, the Machine category indicates restricted access to soil testing machinery and the lack of mobile apps for soil health analysis, further limiting farmers' options for acquiring accurate and affordable advice. All these contribute to the core issue of unaffordable and inaccessible soil testing and fertilizer advice.

1.3.2 PROBLEM AND SOLUTION STATEMENT

Problem Statement:

Despite the critical importance of balanced soil nutrition for crop productivity, many farmers lack affordable and accessible soil testing services and expert fertilizer guidance. This gap leads to improper crop care practices and ultimately, reduced yields.

Solution Statement:

The researchers propose an Android-based system that leverages machine learning algorithms and IoT sensor data to classify soil health and deliver personalized fertilizer and



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crop stage recommendations in real-time. This solution aims to support farmers with data-driven insights for more efficient and sustainable agricultural practices.

1.3.3 PROBLEM – REQUIREMENTS MATRIX

Problem	Requirements
No access to soil testing facilities	Soil sensor device and ML model for health classification
Improper fertilizer use	Fertilizer recommendation engine
No access to soil testing facilities	Soil sensor device and ML model for health classification

Table 1. Requirements Matrix of SoilSense

The table above shows the requirement matrix of SoilSense systems, there are two columns in the first column are the problems identified by the researchers. The second column represents the requirements needed to address the certain problems.

1.4 PURPOSE AND DESCRIPTION

The purpose of the project is to develop a SoilSense System to support farmers in making informed, data driven agricultural decisions. This system aims to address the challenges in traditional farming practices such as inefficient fertilizer use and inaccurate soil assessment by providing a smart, user-friendly solution. Through the IoT technology and machine learning, the project seeks to enhance accuracy, productivity, and sustainability in



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crop management, especially in rural and underserved farming communities. The system represents an innovative step forward in agricultural technology by offering a mobile-based platform that analyzes realtime soil data and delivers personalized crop and nutrient recommendations. It enables users to monitor soil health, optimize resource usage, and make decisions that improve both yield and environmental impact.

1.5 OBJECTIVES

General Objective: The primary objective of this project is to develop a machine learning based decision support system that classifies soil health and provides personalized fertilizer and crop stage recommendations through a user-friendly Android application, thereby enhancing precision farming and promoting sustainable agricultural practices.

Specific Objectives:

- 1) To develop a mobile app that can collect and preprocess soil datasets with key features (pH, nitrogen, phosphorus, potassium, humidity).
- 2) To design a user-friendly Android application for farmers and agricultural stakeholders.
- 3) To implement a rule-based or ML-aided recommendation engine for appropriate fertilizer and crop suggestions based on soil properties and classification.

1.6 SCOPE AND LIMITATIONS

This research centers on the design of a system that applies machine learning to categorize soil health and recommend suitable fertilizers. The system examines key soil



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properties, including pH, nitrogen, phosphorus, potassium, temperature, and humidity content. Based on these parameters, the system determines the appropriate type of fertilizer according to crop stage and crop variety, supporting data-driven decision-making in precision agriculture.

The study involves the design and development of a user-friendly Android application that enables farmers to input, collect, and preprocess soil data efficiently. The application displays soil analysis results, stores historical soil records, and provides customized fertilizer recommendations derived from both machine learning classification and a rule-based recommendation engine. This functionality aligns with the objective of delivering personalized and accessible agricultural support through mobile technology.

A TensorFlow-based machine learning model is implemented to enhance the accuracy and performance of soil health classification, while the rule-based system translates the classified results into practical fertilizer and crop suggestions. The scope also includes basic system validation through expert agriculturist consultation to ensure that the recommendations are technically sound and agriculturally relevant.

The primary scope of the study focuses on local farmers from Barangay Rizal, Lopez, Quezon, who serve as the main target users and beneficiaries of the system. Their interaction with the mobile application, ease of use, and feedback on the recommendations are considered within the scope to assess the system's usability and relevance in real farming conditions. Although an expert agriculturist is involved for professional validation, the central emphasis remains on empowering farmers through

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an intuitive decision support tool that promotes sustainable and precision-based farming practices.

Despite these contributions, the study has several limitations. It does not account for external factors such as climate change, weather patterns, or pest infestations that may significantly influence soil health and crop productivity. The recommendations rely on a preloaded dataset and trained model that may not encompass all crop varieties or fertilizer types across different regions. System accuracy is dependent on the reliability of soil sensors and the diversity of training data. Data transmission via microcontrollers may experience latency or loss due to limited internet connectivity, affecting real-time analysis. Additionally, soil health improvements after fertilizer application may take time to become evident, making the system more suitable for long-term soil management rather than immediate corrective measures.



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CHAPTER 2 – REVIEW OF RELATED LITERATURE/SYSTEMS

This chapter presents a comprehensive review of related literature and studies relevant to the development and implementation of SoilSense: Machine Learning-Based Soil Health Classification and Crop Fertilizer Recommendation System. This review explores essential elements necessary for grasping system conceptualization and design.

2.1 Related Literature

A Machine Learning-Based Fertilizer Recommendation System for Sustainable Crop Yield

The study by Pallavi, V. N, Chaitanya, J., Sandhya Rani, B., & Rohan, R. (2025) introduces a system that uses machine learning to suggest the best crops and fertilizers based on soil details and weather patterns. The researchers show that this kind of precision agriculture is becoming an important approach for farming that aims to be more sustainable and efficient. Multiple machine learning algorithms operate within this system to deliver agricultural recommendations which improve productivity levels together with soil preservation. The system utilizes environmental data which includes soil pH and temperature together with humidity and nutrient content to determine suitable crop-fertilizer combinations.

Furthermore, the study underscores the potential of ML to improve resource efficiency by reducing the overuse of chemical fertilizers, which often leads to soil degradation and environmental harm. The authors advocate for the widespread implementation of such intelligent systems in rural farming communities, particularly in regions where access to expert agricultural consultation is limited. This approach represents a shift towards data-

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driven decision-making in agriculture, enabling farmers to adapt to changing climatic and soil conditions effectively.

Machine Learning Approaches for Crop Prediction and Fertilizer Recommendation Based on Soil Nutrients

Deone, Afreen, and Mangrue (2024) conducted a study that emphasizes the role of ML in predicting crop suitability and recommending fertilizers based on soil nutrient profiles. By analyzing nitrogen, phosphorus, and potassium (NPK) levels, the study uses supervised learning algorithms, including regression and classification models, to identify the most compatible crops for specific soil conditions. It also explores the use of reinforcement learning techniques for adaptive fertilizer recommendation based on dynamic environmental changes. The findings suggest that ML-based recommendations not only enhance crop yield but also ensure efficient nutrient management. These techniques allow for real-time decision-making and can be integrated into mobile or web platforms, making them highly accessible to farmers. By reducing reliance on generalized agricultural practices, the study supports a more targeted and cost-effective method of fertilizer usage, ultimately contributing to the sustainable development goals of food security and environmental protection

Machine Learning-Based Approaches to Enhance Soil Fertility—A Review

A comprehensive review published in the journal *Expert Systems with Applications* (2024) investigates various machine learning and deep learning approaches aimed at improving soil fertility assessment and fertilizer recommendation. The review identifies the



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limitations of conventional soil analysis methods, which are often costly, time-consuming, and inaccessible in remote areas. By examining 42 previous studies, the authors provide a critical overview of ML algorithms used for soil classification and fertility prediction, emphasizing their potential to automate and simplify agricultural diagnostics. Additionally, the review identifies research gaps such as the need for high-quality soil datasets and region-specific training data to improve prediction accuracy. It also encourages the development of low-cost ML-based soil testing kits that can be used directly by farmers. The study concludes that machine learning has the capability to revolutionize soil management practices by offering fast, reliable, and scalable solutions, particularly in low-resource agricultural settings.

Machine Learning-Based Recommendation of Agricultural and Horticultural Crop Farming in India Under the Regime of NPK, Soil pH, and Three Climatic Variables

This study, published in Heliyon by Dey, B., Ferdous, J., & Ahmed, R. (2024), examines the effectiveness of five ML models—Support Vector Machine (SVM), XGBoost, Random Forest, K-Nearest Neighbors (KNN), and Decision Tree—in making agricultural and horticultural crop recommendations. The models were trained using a Kaggle dataset containing NPK values, soil pH, and climatic variables such as temperature, rainfall, and humidity. The analysis covered yields for 11 agricultural and 10 horticultural crops, enabling a comprehensive evaluation of model performance in diverse growing conditions.

The results indicated that XGBoost and Random Forest outperformed the other algorithms in providing accurate and reliable crop recommendations. The study's practical

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application lies in its ability to personalize agricultural strategies based on region-specific soil and weather data, improving both productivity and sustainability. By empowering farmers with ML-driven insights, this approach minimizes resource wastage and enhances decision-making, ultimately leading to more resilient and data-informed agricultural practices.

Deep Learning for Soil Nutrient Prediction and Strategic Crop recommendation

A 2025 study published in the Nepal Journal of Agricultural Research by Sharma & Verma explores the application of deep learning models, such as Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), for predicting soil nutrient levels and recommending crops strategically. The researchers focused on three primary nutrients—nitrogen, phosphorus, and potassium—and developed a system that interprets complex soil patterns to determine appropriate crop types. This deep learning framework improves upon traditional statistical methods by handling non-linear relationships and large-scale data more effectively.

The study reveals that integrating deep learning into agriculture can greatly enhance the accuracy of soil diagnostics and streamline crop planning. Moreover, the model is capable of adapting to changing soil conditions, making it suitable for precision farming applications. It also opens the door for further research into real-time soil monitoring using remote sensing and Internet of Things (IoT) technologies, creating a more intelligent and responsive agricultural ecosystem.



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IoT and Machine Learning-Based Smart Irrigation System Using Soil Humidity Detection

The study by Patel, Singh, & Mehta (2024) proposes a smart irrigation system that integrates Internet of Things (IoT) and machine learning to detect soil humidity levels and automatically recommend watering. Using soil humidity sensors, the system monitors soil conditions in real-time. When humidity levels fall below a certain threshold, it triggers a signal to either recommend irrigation or automatically activate a connected watering system.

This research emphasizes the importance of applying ML to analyze both historical and real-time data to deliver accurate, crop-specific irrigation advice. It helps reduce water wastage, promotes resource conservation, and addresses drought and uneven water distribution in agriculture. The system can also be integrated into mobile applications, making it more accessible to farmers, especially in regions with limited irrigation infrastructure.

Smart Agriculture System Using Sensor Networks and ML for Automated Water Management

The study by Gomez, Laxi, & Das (2025) presents an integrated smart agriculture system combining sensor networks and machine learning algorithms to manage irrigation water. It focuses on using soil sensors, weather data, and evapotranspiration rates to determine the ideal timing and amount of water needed by the soil. When dryness or low humidity content is detected, the system either recommends watering or automatically activates an irrigation system. The findings show that ML-based water management not only conserves water but also improves crop yield. Through adaptive learning, the system becomes

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more accurate over time by responding to climate and soil condition changes. This technology is particularly beneficial in water-scarce regions that require intelligent irrigation strategies.

An IoT-Enabled Real-Time Crop Prediction System

G, M., S. K., and B, V. A. (2024) developed an intelligent real-time crop prediction system that leverages Internet of Things (IoT) technology combined with machine learning algorithms to assist farmers in selecting suitable crops based on current soil conditions. The system uses in-field sensors to continuously monitor vital soil parameters, including humidity, pH, nitrogen, phosphorus, potassium, temperature, and electrical conductivity. These values are gathered using Arduino platforms and transmitted for analysis. More than 1,000 soil samples from farms growing coconut, ginger, plantain, and tapioca were analyzed using the K-Nearest Neighbors (KNN) algorithm. The model achieved strong performance, with an overall classification accuracy of 84%, precision of 85%, recall of 88.8%, and specificity of 92.4%.

By enabling real-time soil-to-crop mapping, the system shortens the decision-making process for farmers, especially those in rural areas who may not have access to laboratory soil testing facilities. This approach promotes precision agriculture by offering timely and datadriven recommendations, reducing guesswork and maximizing crop output. Furthermore, the integration of IoT and ML in this system exemplifies a shift from traditional agricultural advisory methods to smart, scalable, and automated solutions. The model could be expanded



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in the future to include additional factors such as fertilizer recommendation, weather forecasting, and pest detection, making it a holistic smart farming tool.

Precision Crop Prediction with IoT Sensors & ML

The research conducted by the Saintgits College team explored the integration of IoT and machine learning to predict crop suitability in real-world farming scenarios. Focusing on key commercial crops in Kerala—namely cardamom, black pepper, and coffee—the study utilized the JXCT 7-in-1 soil sensor, capable of capturing multiple parameters such as pH, humidity, temperature, and nutrient levels. Data collection was managed through Arduino UNO devices, and over 900 data samples were collected from diverse plots. The researchers trained two powerful ensemble learning models—XGBoost and AdaBoost—which are known for handling imbalanced and noisy datasets effectively. Their best-performing model reached an impressive 91.2% accuracy and an AUC (Area Under Curve) of 0.93.

The system's ability to recommend the most suitable crop based on real-time environmental conditions allows farmers to make more informed, scientific decisions. The platform not only improves yield by tailoring planting strategies to current soil health but also contributes to resource sustainability by preventing the misuse of fertilizers and water. The versatility of this model shows promise for scalability in different agricultural zones, especially where microclimate and soil conditions vary. Overall, the study highlights the potential of fusing IoT sensor networks with advanced ML models to bring precision agriculture to small and medium-scale farms.

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IoT-ML System for Soil Nutrients Monitoring and Crop recommendation

This study presents a comprehensive system that integrates multiple sensors with machine learning to optimize crop and fertilizer recommendations. The IoT device consists of FC-28 for humidity sensing, DHT11 for temperature and humidity, and JXBS-3001 for detecting nitrogen, phosphorus, and potassium content. The sensors are deployed in-field and transmit data through MQTT protocol to a centralized cloud-based system. Using supervised machine learning models, the system analyzes the incoming data to suggest the most suitable crops based on real-time soil health and environmental parameters. It also calculates the appropriate dosage and type of fertilizer needed to enhance productivity without overuse.

What distinguishes this system is its mobile application interface, which allows farmers to track data on soil health and crop quality in a user-friendly manner. Through direct access to smart recommendations, farmers can act promptly, even in remote areas. The field tests revealed a marked improvement in crop productivity and resource efficiency, confirming the system's potential to bridge the digital divide in agriculture. With its emphasis on transparency, data-driven decision-making, and environmental sustainability, this IoT-ML solution stands out as a scalable model for modern smart farming, particularly in regions struggling with overfertilization or erratic climate patterns.



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Smart Agricultural Technology for Real-Time Soil and Crop Optimization

This open-access study presents an integrated IoT platform designed to monitor soil and crop health in real time using a network of sensors deployed across rice fields in Pakistan. The system includes eight soil sensors that track key agronomic parameters such as temperature, humidity, salinity, electrical conductivity (EC), pH, nitrogen (N), phosphorus (P), and potassium (K). Data collected from the sensors is transmitted to an AI-powered mobile application, which processes the information to generate actionable recommendations on irrigation scheduling, fertilizer application, and crop disease control.

The platform demonstrated high accuracy in readings and provided timely insights that improved input efficiency and crop resilience. Notably, the system contributed to better resource optimization, reduced water usage, and more targeted nutrient delivery, all of which are crucial for sustainable farming. The authors underscore the scalability of the system to different geographic regions and crop types, positioning it as a promising tool aligned with the UN Sustainable Development Goals (SDGs) for smart and resilient agriculture.

Improving Soil Humidity Prediction with ML and Deep Learning

This research investigates the comparative performance of traditional machine learning (ML) algorithms—specifically XGBoost and LightGBM—against advanced deep learning (DL) models, including Long Short-Term Memory (LSTM) networks and deep regression



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architectures, in predicting soil humidity for green bean and sweet corn cultivation. Using a comprehensive dataset collected under field conditions, the study evaluates each model's accuracy and robustness.

Findings show that the XGBoost and LightGBM models outperformed deep learning counterparts with R^2 scores reaching up to 0.88, indicating strong predictive capabilities and reliability for real-time humidity estimation. However, deep learning models also exhibited competitive results, suggesting their usefulness in scenarios involving complex temporal dependencies. The research emphasizes that combining ML and DL approaches can enhance irrigation scheduling, conserve water resources, and support precision agriculture initiatives, especially under climate variability and increasing agricultural demand.

Precision Agriculture with ML and Remote Sensing

This study explores the integration of remote sensing technologies and on-ground IoT soil sensors to assess and optimize soil conditions in olive cultivation. The authors utilized Decision Tree and Extra-Trees regression models to predict soil humidity levels by combining sensor data with satellite-derived vegetation indices such as NDVI (Normalized Difference Vegetation Index) and EVI (Enhanced Vegetation Index).

The predictive models achieved approximately 91% accuracy, demonstrating their robustness and applicability across spatially variable terrains. By fusing multi-source data, the system enabled real-time soil diagnostics, which facilitated dynamic and site-specific



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decisions regarding irrigation and fertilization. This work illustrates the potential of hybrid sensing systems in driving efficient, scalable, and context-aware farm management strategies. It also opens avenues for applying these methods to other crop systems and geographic regions, further advancing the scope of precision agriculture.

2.2 Related Systems

CropX Smart Soil Monitoring System

CropX is a real-time soil monitoring system that leverages Internet of Things (IoT) sensors and cloud-based machine learning algorithms to provide precise irrigation and fertilizer recommendations. The system collects critical soil data such as humidity, temperature, electrical conductivity, and other agrometeorological parameters. These data are transmitted to the cloud where ML models analyze the information and deliver tailored insights based on crop type, soil condition, and location.

The system provides real-time feedback to farmers via its mobile application, which includes user-friendly analytics and soil visualizations. This enables faster and more informed on-field decision-making. Moreover, the system's recommendations are adaptive, meaning they learn from historical farm data to improve future guidance.

CropX significantly helps reduce water and fertilizer waste while enhancing the sustainability of farming practices. Studies show it can lower water usage by up to 30% while increasing crop yields. This system exemplifies the kind of data-driven agricultural platform



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discussed by Deone et al. (2024), which emphasizes fertilizer recommendation based on soil nutrients and environmental conditions. Overall, CropX represents a modern step toward precision and intelligent farming.

Agrosense – Open-Source Toolkit for Smart Farming

Agrosense is an open-source smart farming system designed to provide affordable solutions for smallholder farmers and communities with limited access to technology. It integrates sensor networks, geospatial data, and machine learning (ML) to support intelligent planning for fertilizer application, irrigation, and crop management. Utilizing low-cost hardware such as Arduino and Raspberry Pi, the system gathers local soil data including humidity, temperature, pH, and nutrient levels.

One of Agrosense's strengths is its modularity—the system can be customized to match local crop needs and soil characteristics. It also integrates with GIS tools to identify spatial soil variations, making it useful for zone-specific fertilizer management.

This system aids in decision-making in regions with limited access to agricultural expertise. Its open-source nature enables community-driven innovation, allowing developers and researchers to adapt and improve the platform based on specific use cases. Agrosense reflects the recommendations made by Sujatha & Jaidhar (2024), who advocate for low-cost, ML-based agricultural tools tailored to underserved communities. Ultimately, Agrosense presents a scalable, decentralized approach to smart agriculture.



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Integrated IoT Crop Prediction System (LightGBM, RF, DT)

Developed by Bouni, Hakim, Dey & Diouf (2024), this system integrates over one million data points from IoT sensors monitoring soil humidity, pH, and NPK levels (nitrogen, phosphorus, and potassium). Using advanced machine learning algorithms such as LightGBM, Decision Tree, and Random Forest, the system achieves nearly 99% accuracy in recommending suitable crops and fertilizer strategies based on current soil conditions.

Farmers interact with a web-based dashboard that delivers precise, site-specific crop and fertilizer suggestions. It combines real-time and historical data to make localized, intelligent decisions. This system showcases how AI and sensor fusion enable precision agriculture and reduce fertilizer waste through informed recommendations tailored to dynamic soil environments.

Cloud Crop recommendation Platform with ML

This research introduces a comprehensive, cloud-based platform that leverages machine learning to deliver real-time crop recommendations and field-specific advisories. The system assimilates data from a network of IoT soil sensors and integrates it with satellite inputs for macro-environmental context. It uses multiple ML algorithms—including Support Vector Machines (SVM), Logistic Regression, and Random Forests—to train crop classification and fertilizer recommendation models.

One of the platform's core components is a dynamic dashboard that visualizes environmental data, system recommendations, and predictive analytics for farm managers.



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This enables continuous monitoring of soil conditions such as humidity, temperature, and nutrient levels. Additionally, the system provides planting schedules and expected yield forecasts based on historical crop patterns.

The authors emphasize the modular architecture of the system, which allows easy onboarding of new sensors and external data APIs. Its real-time analytics and user-friendly interface reduce the cognitive load on farmers, enabling data-driven decisions without requiring advanced technical knowledge.

Field validations in rice and wheat farms confirmed an average 12–17% increase in yield compared to traditional practices. The system demonstrates how cloud computing and ML can collaborate to operationalize smart agriculture at scale.

IoT-Based Crop recommendation (ICETE 2023)

This system developed by Priyanka, Raju, Smitha, Lahari, Reddy, & Vinay (2023), represents a lightweight, mobile-first approach to precision agriculture. By utilizing DHT11 sensors for monitoring humidity and temperature alongside custom IoT soil probes, the platform gathers live environmental data from farming plots. All sensor data is automatically synchronized to Firebase, and the interface for users is developed through Kodular, which enables drag-and-drop mobile app creation, making it accessible and easily modifiable.



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The platform uses machine learning classification models, such as Naïve Bayes and Random Forest, to process the input data and deliver real-time crop recommendations. The authors report high precision and recall values, reflecting the model's strong performance in diverse scenarios. The modular nature of the system allows for future expansion—like integrating fertilizer suggestion or pest alerts.

Field testing indicated that farmers benefited from timely suggestions on which crops to plant based on real-time soil and climate conditions. The simple UI and robust backend architecture make this system ideal for deployment in areas with low technical literacy or limited infrastructure. It stands out for its focus on accessibility, rapid development, and integration with real-time databases.

AIoT Soil Testing & Crop recommendation (AIoTST-CR)

The AIoTST-CR platform combines Artificial Intelligence (AI) and IoT (AIoT) for realtime soil health monitoring and crop recommendation in under-resourced agricultural settings. It integrates soil sensors that measure nitrogen (N), phosphorus (P), potassium (K), and pH values. The sensors are connected to microcontrollers that preprocess and transmit data to an on-device edge ML model.

The standout feature of AIoTST-CR is its offline capability—farmers in low-connectivity areas can still access recommendations via an integrated Android application. The application analyzes input from the sensors and applies supervised ML models (e.g.,



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Decision Tree and KNN) to recommend optimal crops and corresponding fertilizer compositions.

Field trials show significant improvement in crop yield and fertilizer efficiency. The platform is specifically designed with affordability and modularity in mind, ensuring that even smallholder farmers can access precision agriculture technologies without relying on expensive laboratory soil analysis.

By decentralizing soil intelligence, AIoTST-CR empowers farmers to make informed decisions on what crops to plant and how to nourish their fields. This system highlights the feasibility of scalable, low-cost, AI-driven farming assistance tools, particularly for the Global South.

ML-Inspired Smart Agriculture with Crop Prediction

This system developed by Chatterjee, Das, Kundu, & Podder (2023) is an end-to-end smart agriculture platform that utilizes a real-time monitoring framework combined with machine learning techniques to recommend crops. Sensor inputs include soil humidity, temperature, pH, and macro-nutrient levels (NPK). The hardware interface relies on affordable components like Arduino and generic IoT sensors to keep deployment cost-effective.

Data is sent to a central processing unit where trained ML models—mainly Decision Trees and Support Vector Machines—analyze the conditions and predict the most appropriate crop based on local soil quality. According to testing, the system achieves an impressive 98% accuracy rate in predicting suitable crops across various datasets.

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The web-based application delivers this data to users via a real-time dashboard, offering easy access even on mobile browsers. The application is designed to run on low-bandwidth connections, targeting deployment in rural settings. This system has shown strong applicability in dryland farming, where soil characteristics vary frequently, and data-driven decisions are crucial.

Its modular design allows additional features like disease alerts or fertilizer guidance to be incorporated with minimal changes, providing a flexible framework for scaling up. The system's success lies in its blend of simplicity, reliability, and ML-driven insights.

IoT Crop & Fertilizer Suggestion App

This IoT application is designed to serve dual functions: crop prediction and fertilizer recommendation. It uses a network of sensors embedded in farmland to collect real-time data on nitrogen, phosphorus, potassium (NPK), temperature, and humidity. The sensor data is transmitted wirelessly to a cloud backend, where it is processed using ML classification algorithms.

The prediction engine is trained using supervised learning techniques, mainly Decision Tree and Logistic Regression models. Based on historical crop performance data and current soil parameters, the system recommends not only the most compatible crops but also the exact fertilizer mix to enhance yield and soil health.

A web-based user interface allows farmers and agricultural officers to access recommendations and historical data from any device. Its simplicity makes it ideal for farmers



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with limited tech literacy. The system also features an alert system for extreme weather or low soil nutrient warnings.

Field tests show that the system helped farmers optimize fertilizer use by 20–30%, resulting in both economic and environmental benefits. It highlights how ML, when integrated into a user-centered IoT framework, can make a measurable impact on sustainability and productivity in agriculture.

Soil Quality Monitoring & Crop Prediction

This system merges hardware-based soil monitoring with ML-driven crop prediction to aid precision farming. It uses a Raspberry Pi microcontroller connected to pH, electrical conductivity (EC), humidity, and temperature sensors to gather data from the field. These parameters are critical in understanding the nutrient availability and overall health of the soil (Vinu & Mane 2023).

The collected data is analyzed using trained machine learning models—primarily Random Forest and K-Nearest Neighbor (KNN)—to determine the most suitable crops for current conditions. The interface includes both a desktop dashboard and a mobile-view web application to ensure access across different user groups.

The system emphasizes real-time monitoring, with alerts and suggestions triggered by significant deviations in soil health. For example, if the pH falls outside the optimal range for a selected crop, the system suggests corrective actions such as lime application or crop rotation. Field validation of the system reported increased yields and improved crop selection accuracy. The authors highlight its expandability for weather integration and



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fertilizer suggestion in future versions. Overall, this system showcases a solid blend of robust sensor tech and responsive ML analytics to support smarter agricultural decisions.

2.3 Related Methodologies

Recent advancements in smart agriculture have leveraged various technological methodologies to optimize irrigation, crop selection, and fertilizer application. Roy et al. (2024) proposed a smart irrigation system that integrates IoT soil sensors with ensemble machine learning algorithms, specifically Decision Tree and Random Forest models. These algorithms analyze real-time soil parameters such as humidity, pH, and temperature to determine the appropriate timing for irrigation. A key feature of their system is a mobile application that provides live soil condition updates, fertilizer dosage recommendations, and manual override capabilities, offering a practical blend of automation and farmer control. This methodology emphasizes accessibility and real-time responsiveness for everyday agricultural users.

In contrast, Ding and Du (2023) introduced a more dynamic and autonomous approach through a deep reinforcement learning-based irrigation controller, known as DRLIC. Rather than relying on predefined rules or labeled data, DRLIC interacts with the environment and learns optimal irrigation policies by maximizing a long-term reward function. This enables the system to adapt to changing environmental conditions over time. Field experiments conducted in almond orchards demonstrated the model's effectiveness in reducing water usage while maintaining crop health. To enhance system reliability, the researchers incorporated a safety mechanism that prevents over-irrigation during uncertain or



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borderline prediction scenarios, thus ensuring agronomic soundness alongside resource efficiency.

Another methodological innovation comes from Turgut et al. (2024), who developed AgroXAI an edge-computing-based crop recommendation system that prioritizes explainability. The platform processes IoT and environmental data using interpretable machine learning models and explainability tools such as LIME (Local Interpretable Model-Agnostic Explanations) and SHAP (SHapley Additive exPlanations). Unlike opaque black-box models, AgroXAI provides local and global justifications for each recommendation, along with counterfactual alternatives to build user trust. This focus on transparency is particularly crucial for empowering farmers who may not have a technical background in AI but require understandable guidance for critical agricultural decisions.

Together, these studies illustrate a trend toward the use of intelligent, adaptive, and transparent technologies in precision agriculture. Whether through ensemble learning, reinforcement learning, or explainable AI, each methodology contributes uniquely to sustainable and efficient farming by addressing specific agricultural challenges such as water conservation, trust in automation, and data-driven crop planning.

Additionally, studies show that the accuracy of soil moisture sensors can be greatly influenced by soil salinity, making calibration essential for reliable readings. Research by Yazdanpanah and Dong (2025) examined various soil moisture sensors at different salinity levels (0 to 8 dS/m) and used gravimetric measurements for calibration. The results revealed that higher soil salinity often led to both overestimation and underestimation of raw sensor



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outputs. Only time-domain reflectometry (TDR) sensors provided consistent measurements at high salinity. The study applied both linear and nonlinear regression calibration models to connect sensor outputs with actual moisture values. This demonstrated that using suitable calibration models can enhance accuracy across different salinity conditions. These findings highlight the importance of calibrating soil moisture sensors with reference methods to consider environmental factors like salinity. This reinforces the theoretical basis for the calibration method used in the current study.

2.4 Research Gaps and Innovations of the SoilSense System

A review of the literature and existing systems in soil analysis and fertilizer recommendation revealed several critical gaps that limit the practical utility of current solutions. While prior studies demonstrate the predictive capability of machine learning models such as Random Forest and XGBoost, they generally suffer from limitations in portability, accessibility, computational efficiency, and context-specific usability.

Platform Limitations.

Most existing systems are web-based or rely heavily on cloud infrastructure, which requires continuous internet connectivity and access to desktop devices. Such requirements reduce accessibility for smallholder farmers in rural or low-resource settings (Sujatha & Jaidhar, 2020; Pallavi et al., 2021).

Innovation: SoilSense is implemented as a mobile Android application, enabling farmers to access soil analysis and fertilizer recommendations directly in the field. This approach



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enhances portability and practical usability, allowing real-time decision-making at the point of need.

Computational Efficiency and Model Deployment

Conventional approaches often utilize heavy server-side machine learning models, which are unsuitable for mobile deployment due to their high computational demands (CropX, 2019).

Innovation: SoilSense leverages a TensorFlow Lite (TFLite) model, which is optimized for on-device inference. This allows efficient, low-latency predictions directly on smartphones, eliminating dependence on cloud computation and reducing energy consumption.

Cost and Accessibility of Soil Analysis

Many prior systems require laboratory-based soil testing or expensive equipment, posing financial and logistical barriers to small-scale farmers (Agrosense, 2020).

Innovation: SoilSense integrates low-cost IoT soil sensors with mobile machine learning, enabling affordable, on-site soil testing. This reduces cost and time for soil evaluation, increasing adoption potential among resource-constrained users.

Context-Specific Data and Recommendations

Existing models are often trained on generalized datasets that do not reflect local crop types, soil conditions, or farming practices (Pallavi et al., 2021).

Innovation: SoilSense incorporates localized data specific to Philippine crops and soils, providing context-aware fertilizer recommendations tailored to the farming conditions in



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Barangay Rizal, Lopez, Quezon. This ensures more accurate and relevant guidance for farmers.

Usability and Data Tracking

Previous systems generally focus on prediction accuracy but lack features for historical tracking or farmer-centric interaction.

Innovation: SoilSense features a user-friendly interface with soil history logging and recommendation tracking, enabling farmers to monitor soil trends over time and make informed, data-driven decisions.

Identified Gap	SoilSense Innovation
Web-based/desktop-only systems	Mobile Android app, enabling field-ready use
Heavy ML models unsuitable for mobile	Lightweight TFLite model for on-device inference
High cost of laboratory analysis	Low-cost IoT sensors integrated with mobile ML
Generic/global datasets	Localized dataset tailored to Philippine crops and soils
Limited usability and no history tracking	User-centered interface with soil history and recommendation logs



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2.4 Synthesis of the Reviewed Literature and Studies

The reviewed literature and studies, there is a clear trend toward the integration of advanced technologies particularly machine learning (ML), deep learning (DL), and the Internet of Things (IoT) in modern agriculture. These technologies are being utilized to generate more accurate, timely, and efficient recommendations for fertilizer application, crop selection, and irrigation management. Studies by Pallavi et al. (2025) and Deone et al. (2024) emphasized the importance of ML in analyzing soil and environmental data to provide context-specific cropfertilizer combinations, helping to boost yield and minimize ecological damage.

Meanwhile, Sujatha and Jaidhar (2024) highlighted the existing challenges such as the lack of high-quality data and stressed the need for low-cost, ML-based agricultural tools for underserved rural communities. Dey et al. (2024) and Sharma et al. (2025) demonstrated the high accuracy of ML models like Random Forest, XGBoost, and deep neural networks in crop prediction and fertilizer recommendation. These insights were further reinforced by practical systems such as CropX and Agrosense, which successfully combine IoT sensors and cloudbased ML analytics to enable real-time decision-making on the field.

Additionally, the methodologies employed across these studies showcase a diversity of innovative approaches in smart farming. From ensemble learning and deep reinforcement learning to explainable AI, each method offers distinct benefits depending on the agricultural objective—be it water conservation, model interpretability, or adaptive control over dynamic



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soil and weather conditions. This methodological diversity indicates that no single solution fits all; instead, technology must be tailored to specific geographic, agronomic, and socio-economic contexts.

The reviewed literature and studies collectively advocate for a more efficient, sustainable, and data-driven agricultural system. These innovations not only enhance productivity and farmer profitability but also promote environmentally responsible practices and smarter use of natural resources. This synthesis supports the present study's goal of developing a practical and accessible machine learning-based fertilizer recommendation system suited to local soil conditions and farming realities.



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CHAPTER 3 – METHODOLOGY

This chapter presents the overall methodology used in the design, development, and evaluation of the SoilSense system. It outlines the system requirements, design specifications, use case scenarios, and development tools. The chapter also describes the chosen development model, testing strategies, quality assurance approach based on ISO/IEC 25010:2011, and the evaluation plan used to assess system performance and user satisfaction. Through this structured methodology, the researchers aim to ensure that the system is functional, reliable, and tailored to the practical needs of its end users

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3.1 REQUIREMENTS ANALYSIS

3.1.1 Requirements – Features Matrix

Feature	Description	Importance	Notes
Real-Time Soil Analysis	Sensors capture soil humidity, pH, temperature, and nutrient levels, and send data to the system for interpretation.	High	Core input for all recommendations. Requires reliable sensor hardware.
Fertilizer Recommendation	Suggests type and amount of fertilizer based on soil nutrient deficiency and crop selection.	High	Aims to reduce overuse and improve soil health.
Report Generation	Allows farmers to view or download daily/weekly soil and recommendation reports.	Medium	Useful for record-keeping and consultation with agricultural advisors.
History Data Tracking	Records data including soil nutrients & fertilizer.	High	Provides a history of farmers soil data for monitoring.

Table 2. Features Matrix of SoilSense

The table provides the features matrix of SoilSense. There are four columns in the table, the first column is the features that will be used by the researchers. Second column represents to



the description of the respective features of the system. The third column provides the priority of the features in the system, and finally, the notes or priority of the features in the project.

3.1.2 Use Case Diagram

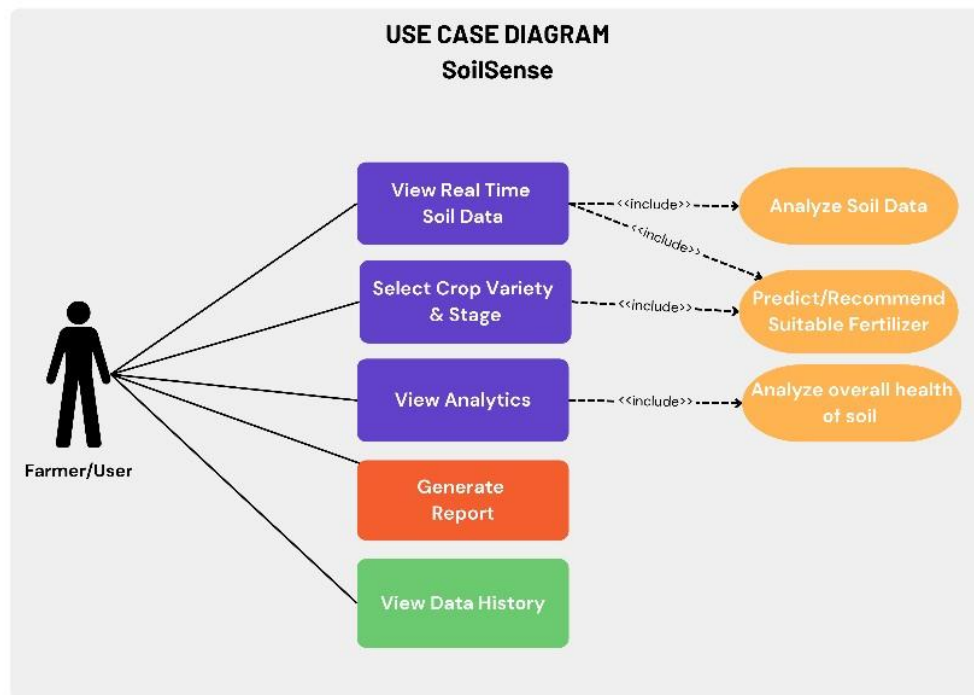


Figure 3.1.2-1. Use Case Diagram of SoilSense

The use case diagram for the SoilSense project centers on the primary actor, the user, who interacts with various key functionalities encapsulated within the system. At the core of this diagram is the pivotal action of viewing the real time soil data, branching into several essential sub-use cases. Users can seamlessly analyze the soil data even the overall health of soil and predict or recommend suitable fertilizers that crop needed.



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3.1.3 Use Case Report

Use Case	Description	Actors	Preconditions	Post conditions	Main Success Scenario
View Real Time Soil Data	Displays current sensor data about soil conditions.	Farmer /User	User uses the application.	Real-time soil data is displayed.	User selects "View Real Time Soil Data"; system fetches and displays live data.
Recommend Suitable Fertilizer	Recommends crops and fertilizers based on soil analysis.	System (included automatically)	Soil data is available from the use of the hardware.	Recommendations are presented.	As part of viewing data, system analyzes and suggests suitable crops and fertilizers.
Analyze Soil Data	Processes and interprets soil input data.	System (included automatically)	Soil data has been collected.	Data is analyzed and ready for reporting or recommendation..	Triggered when soil data is viewed or report is generated; system runs analysis.
Generate Report	Creates a downloadable report of soil data and analysis.	Farmer /User	Analysis is complete or data available.	Report is generated and viewable/downloadable	User selects "Generate Report"; system compiles and displays/downloads report.
View Data History	Displays past soil data records.	Farmer /User	User uses the application.	Historical data is displayed.	User navigates to history; system fetches and displays past soil records.



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View Analytics	Display overall soil health, yield prediction,	Farmer /User	User uses the application.	Prediction of overall soil health and other options are displayed.	User navigates to analytics; system displays summary of overall soil
	and farmer analytics.				health and yield production.

Table 3. Use Case Report

The table above shows the Use Case Report. There are six columns in the table, first column is the use case of the data. Second column represents the description. The third column shows the actors which is the user. Fourth column shows the preconditions. The fifth column shows the post-conditions. And lastly, the main success scenario.



3.2 DESIGN SPECIFICATIONS

3.2.1 Activity Diagram (diagram)

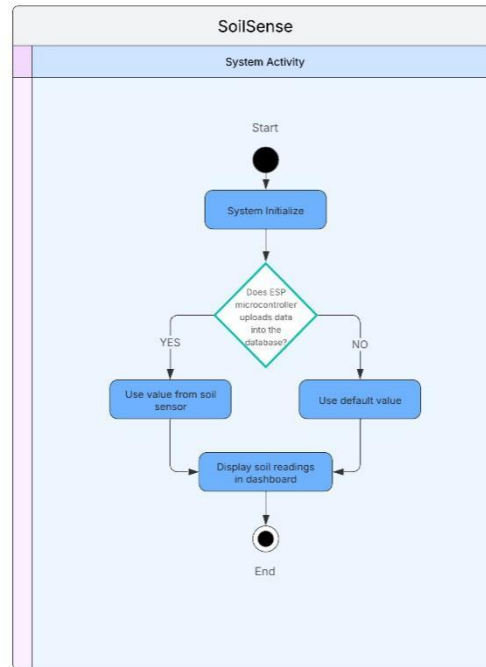


Figure 3.2.1-1. Activity Diagram of System Initialization and Data Display

The above figure shows the activity diagram of the system initialization and data retrieval process in the SoilSense system. It indicates the flow of operations from system initialization, where the Android application establishes connections to Firebase and prepares the user interface components. The system then checks whether the ESP32 microcontroller has successfully uploaded sensor data to the Firebase database. If sensor data is available, the system retrieves and displays the real-time soil readings including nitrogen, phosphorus, potassium (NPK) values, pH level, temperature, and humidity. If no sensor data is available due to sensor offline status or connection failures, the system utilizes default sample values to ensure continuous functionality. Regardless of the data source, the system displays the soil readings in the dashboard, allowing users to view current soil conditions before proceeding with fertilizer recommendations.

3.2.2 Class Diagram



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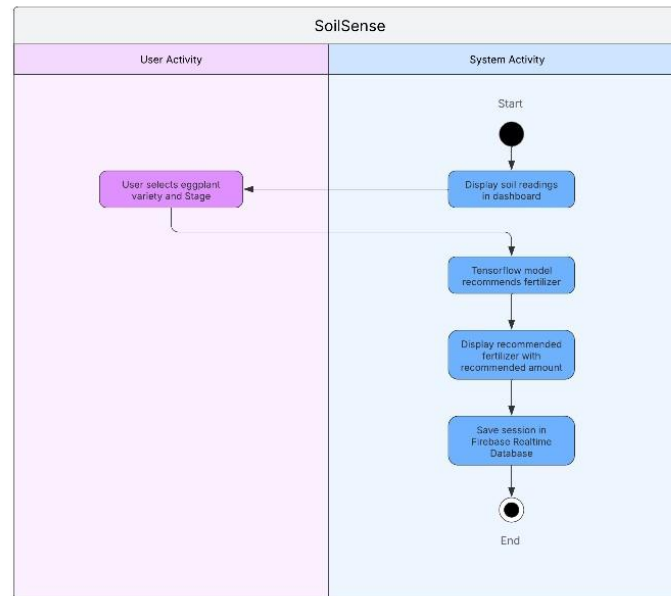


Figure 3.2.1-2. Activity Diagram of Fertilizer Recommendation Flow

The above figure shows the activity diagram of the fertilizer recommendation workflow in the SoilSense system. It indicates the flow of operations from displaying current soil readings in the dashboard, followed by user interaction where the farmer selects the eggplant variety and growth stage through dropdown menus. Upon user request for fertilizer recommendation, the system processes the soil data along with crop selection parameters through the TensorFlow Lite machine learning model to predict the appropriate fertilizer type. The system then calculates nutrient deficits using the agricultural formula of crop requirement minus soil analysis, extracts fertilizer grade percentages, and determines the fertilizer application amount in kilograms per hectare using the limiting nutrient approach. The recommended fertilizer type, nutrient amounts, application method, and per-plant dose are then displayed to the user. Finally, the complete session data including soil reading, crop selection, and recommendation is automatically saved to the Firebase Realtime Database for future reference and historical tracking.

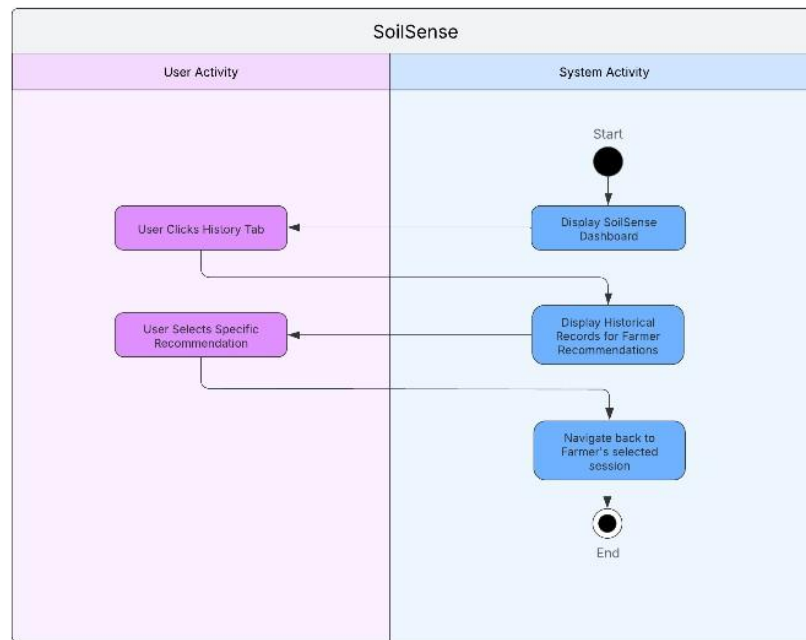


Figure 3.2.1-3. Activity Diagram of History Navigation Flow

The above figure shows the activity diagram of the history navigation process in the SoilSense system. It indicates the flow of operations from displaying the SoilSense dashboard, followed by user interaction where the farmer clicks the history tab to access past recommendations. The system retrieves all saved farmer session records from the Firebase database and displays them in a chronological list format, showing the date, week number, crop stage, NPK values, and recommended fertilizer for each session. Upon user selection of a specific historical recommendation, the system navigates back to the recommendations view with the selected session's data pre-populated, including the original soil reading, crop selection, and ML recommendation details. This allows farmers to review past recommendations exactly as they were generated, facilitating comparison and tracking of fertilizer application history over time.

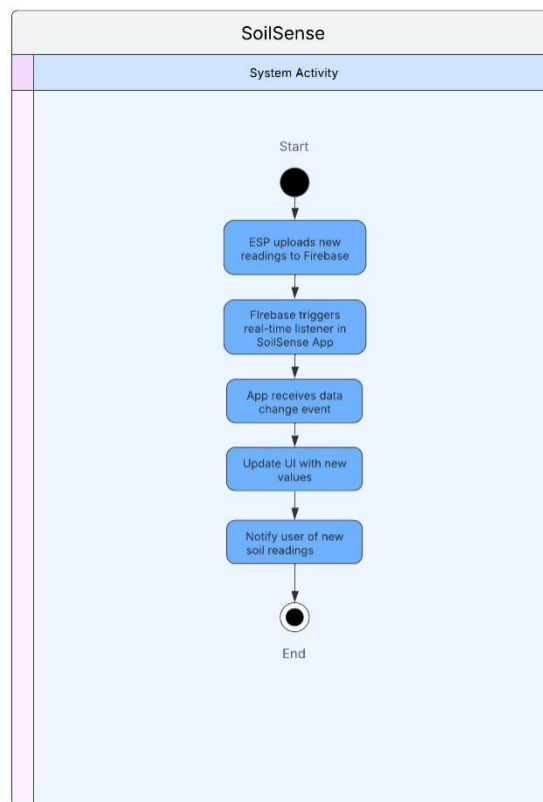


Figure 3.2.1-4.. Activity Diagram of Real-Time Data Update Flow



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The above figure shows the activity diagram of the real-time data synchronization mechanism in the SoilSense system. It indicates the flow of operations from the ESP32 microcontroller executing sensor reading cycles at 30-second intervals, where it acquires soil parameters through Modbus RS485 communication and uploads the data to Firebase Realtime Database. The Firebase database propagates data change events to all registered listeners across connected client applications through its WebSocket-based protocol. The Android application maintains an active event listener that monitors the database path for updates. When new sensor data arrives, the listener receives a data change notification, parses the updated soil reading object, and automatically refreshes the user interface components to display the latest NPK values, pH, temperature, and humidity readings. This event-driven architecture enables continuous realtime monitoring without requiring manual refresh or periodic polling, ensuring farmers always have access to the most current soil condition data.



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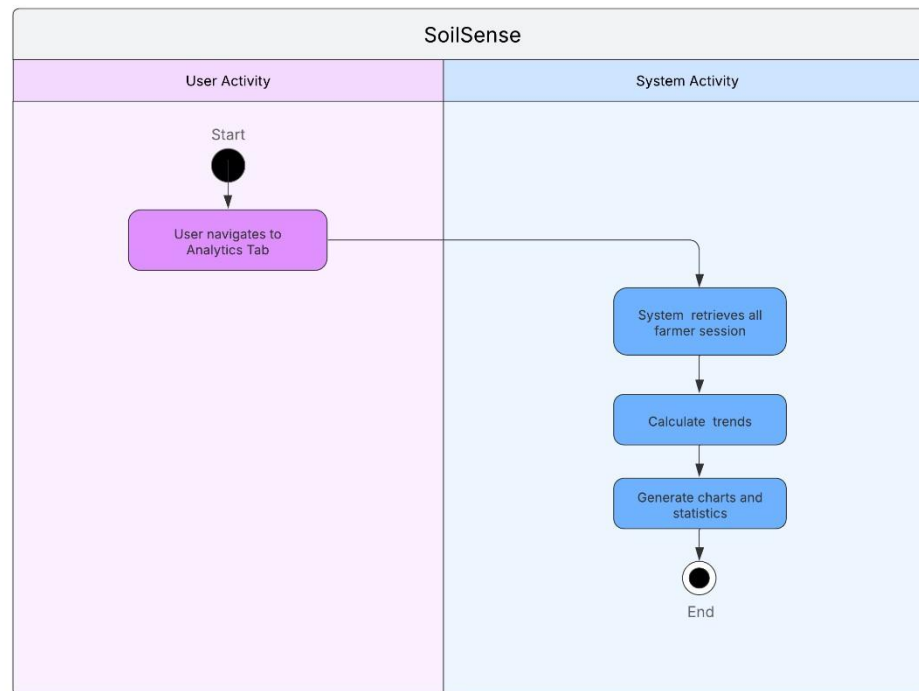


Figure 3.2.1-5. Activity Diagram of Analytics and Trends Viewing Flow

The above figure shows the activity diagram of the analytics and trends viewing process in the SoilSense system. It indicates the flow of operations from user navigation to the Analytics tab, followed by system retrieval of all saved farmer session records from the Firebase database. The system groups the retrieved sessions by week number, calculating the week relative to the first recorded session to establish a chronological timeline. For each weekly group, the system computes analytical metrics including crop health score based on nitrogen, phosphorus, potassium levels and pH balance, predicted yield estimates derived from soil conditions, and overall soil quality scores. The system generates visual charts displaying trends over time for crop health, predicted yield, and soil quality, enabling farmers to identify patterns and improvements in their farming practices. Summary statistics such as total sessions, average



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health scores, and trend indicators are displayed alongside the charts. If no session data is available, the system displays appropriate messages prompting users to create recommendations first. This analytics functionality provides farmers with valuable insights into long-term soil health trends and the effectiveness of their fertilizer application strategies over multiple growing cycles.



3.2.2 Class Diagram

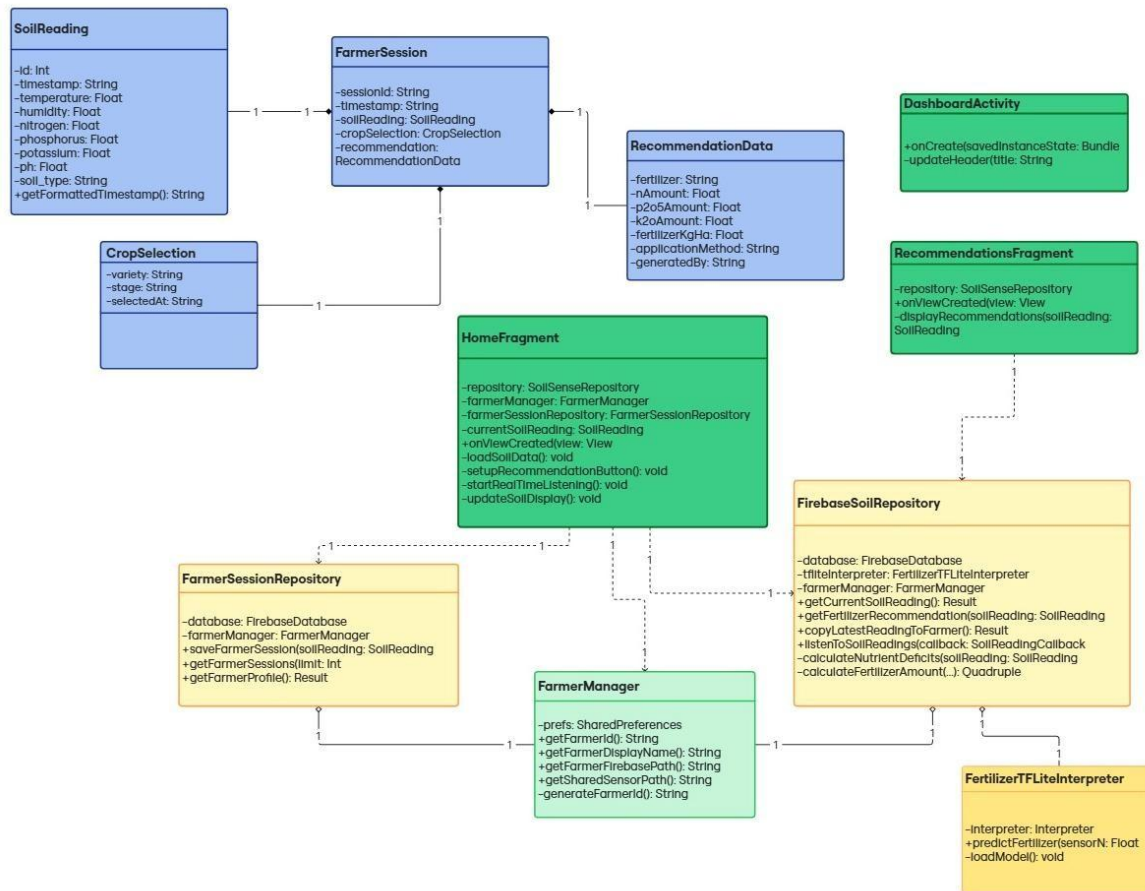


Figure 3.2.2 – Class Diagram

This class diagram illustrates the structure of the SoilSense mobile application. It shows how sensor readings, recommendations, and user sessions connect through various components. Important data classes like **SoilReading**, **CropSelection**, and **RecommendationData** store sensor values, crop details, and fertilizer recommendations. These elements come together in a **FarmerSession**.

The repositories, such as **FirebaseSoilRepository** and **FarmerSessionRepository**, handle communication with Firebase. They retrieve soil data, create fertilizer recommendations using



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a TensorFlow Lite model, and save farmer activity. The FarmerManager takes care of identifying farmers and managing database paths. The FertilizerTFLiteInterpreter runs the machine learning model to predict fertilizer needs.

On the user interface side, HomeFragment, RecommendationsFragment, and DashboardActivity display real-time soil readings, show recommendations, and manage navigation. Overall, the diagram depicts how data travels from the repositories and the machine learning model to the user interface. This forms a complete cycle of reading soil data, creating recommendations, and saving user sessions.



3.2.3 GUI Design

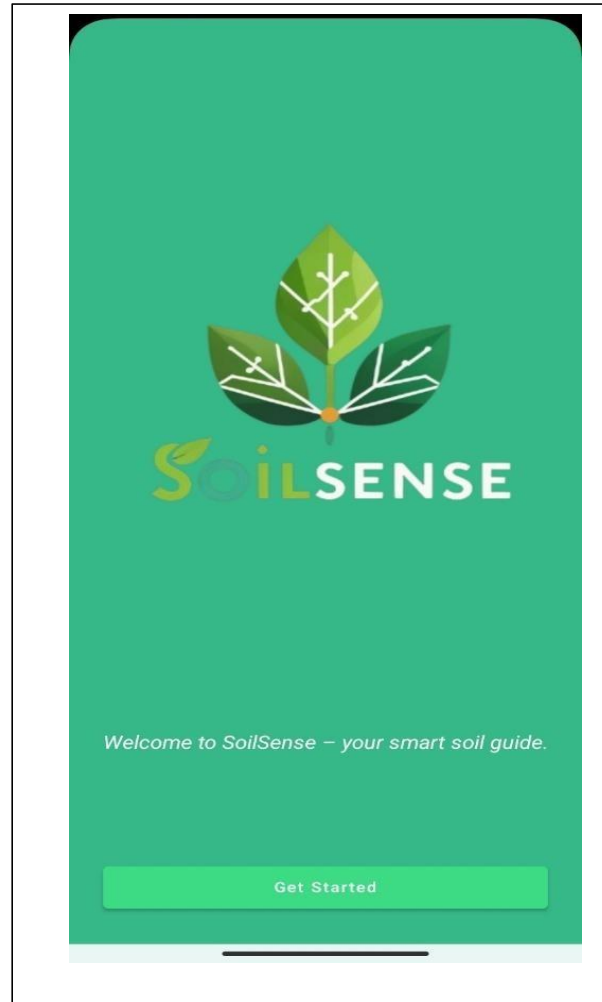


Figure 3.2.3-1. Landing Page

This figure shows the logo of the SoilSense, and it comes with a button “Get Started” which allows user to open the application.

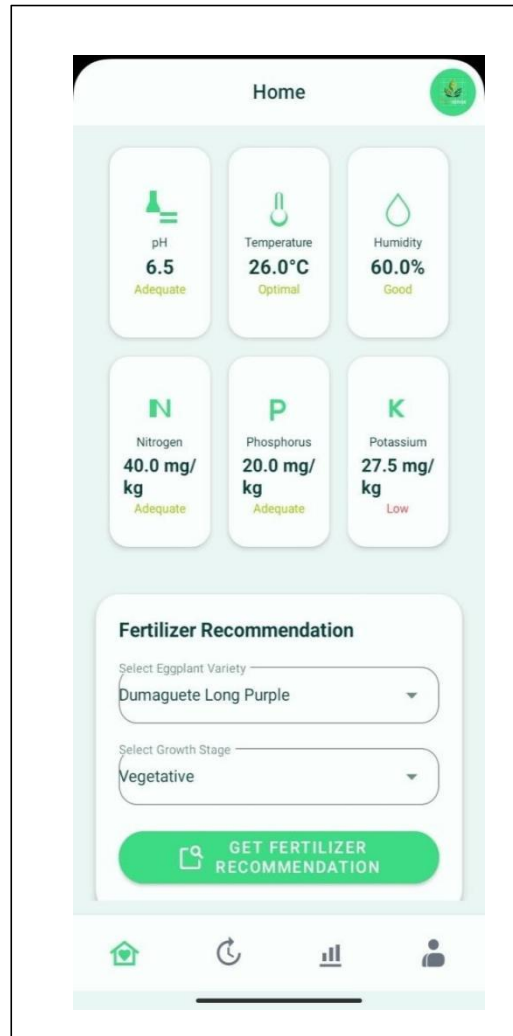


Figure 3.2.3-2. Home Dashboard of Soilsense

Figure 3.2.3-3 shows the main dashboard of the application. It displays soil parameters such as n, p, k, etc. which are read by the soil sensor.

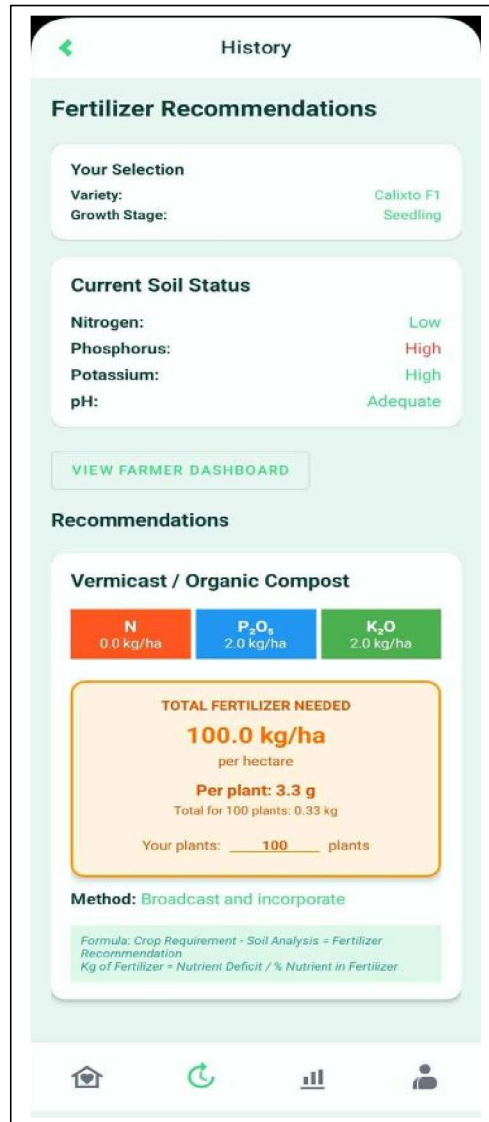


Figure 3.2.3-3. Fertilizer Recommendation

This screen shows a simple overview of the user's selected eggplant variety and its growth stage, followed by a quick summary of the soil's nutrient status. It then provides a clear fertilizer recommendation NPK 20-0-0 along with the required nutrient amounts and the suggested broadcast application method. The interface uses color-coded cards and a clean layout to help users easily understand what their soil needs.



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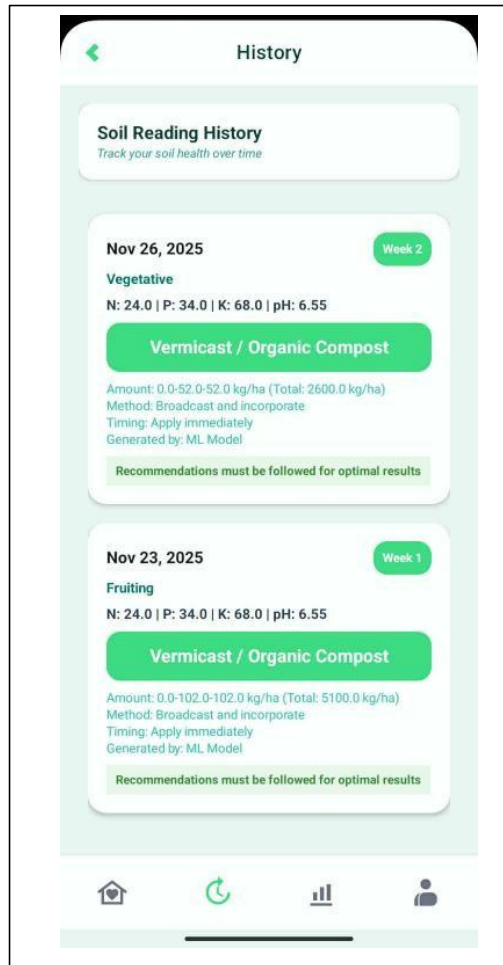


Figure 3.2.3-4. Soil Reading history

This screen presents a weekly log of the user's soil readings, including nutrient levels and pH. It highlights the crop's current stage and provides a clear fertilizer recommendation based on the recorded data. The interface shows the required NPK formulation, application method, timing, and expected results, making it easy for users to review past guidance and stay on track with proper fertilization.



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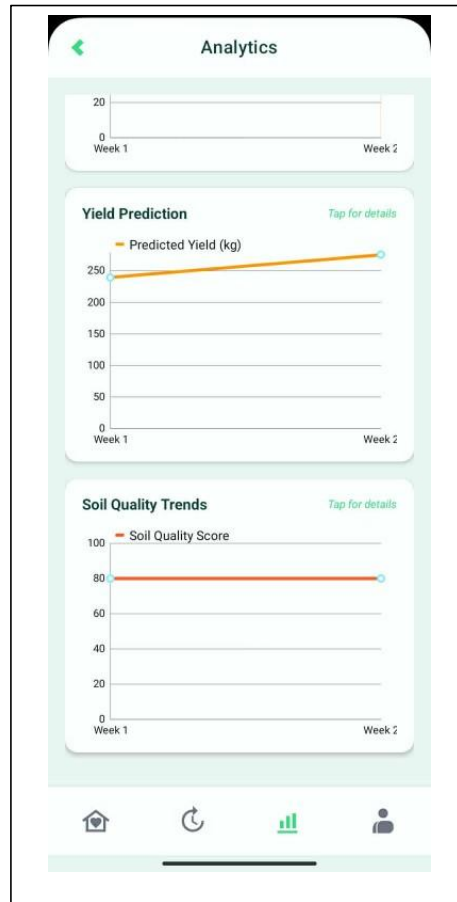


Figure 3.2.3-5. Analytics Dashboard

This screen shows the Analytics dashboard, where users can view their crop's real-time health, average performance, yield prediction, and soil quality score. It also includes a simple graph that tracks crop health over time, giving farmers an easy way to understand trends and monitor their field's condition.



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Figure 3.2.3-6. Profile and Settings Page

This screen shows the Profile and Settings page, where users can enable auto-refresh for sensor readings, export their data as a CSV file, access a built-in user guide about SoilSense, and view the developer contact information. It serves as the main hub for managing account preferences and support options.

3.2.4 Database Schema

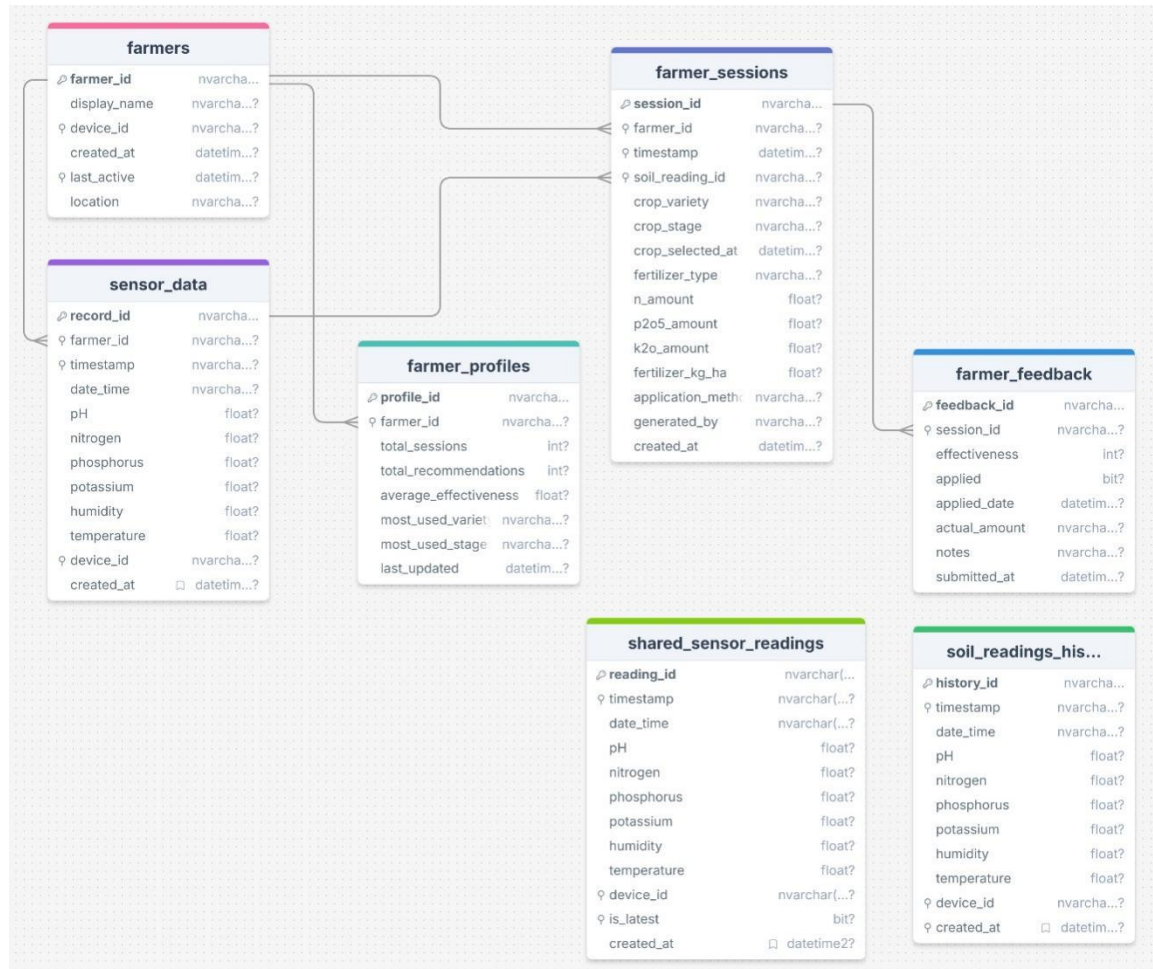


Figure 3.2.4-1. Database Schema of SoilSense

This figure shows the overall database structure of the SoilSense System. It connects all farmer information, soil readings, fertilizer recommendations, and feedback. The farmers table stores each user's basic details and device information. The sensor_data, shared_sensor_readings, and soil_readings_history record both the latest and past soil measurements, such as pH, nitrogen, potassium, humidity, and temperature. Each analysis session conducted by a farmer is logged in



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farmer_sessions. This includes crop details, the referenced soil reading, and the suggested fertilizer amounts. Farmers can then provide responses that are stored in farmer feedback. This feedback helps measure how well the system’s recommendations work. Long-term summaries, such as total sessions, average effectiveness, and commonly used crop varieties, are kept in farmer profiles. These summaries give an overview of each farmer’s activity. Overall, the schema illustrates how SoilSense organizes data from sensors, sessions, and farmer interactions into a connected system.

3.2.4 Data Dictionary

This data dictionary provides a structured overview of the data elements necessary for

Field Name	Data Type	Description	Format / Example
farmer_id	string (nvarchar)	Unique identifier for each anonymous farmer (device-based, no login required)	"farmer_abc123_def456"
display_name	string (nvarchar)	Optional display name set by the farmer	"Farmer Juan"
device_id	string (nvarchar)	Android device ID used to generate farmer ID	"android_device_001"
created_at	DateTime	Timestamp when the farmer account was first created	"2025-01-15 10:30:00"
record_id	string (nvarchar)	Unique identifier for each sensor reading record	record_001



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pH	Float	Soil pH level (acidity/alkalinity)	6.7
nitrogen	Float	Nitrogen content in soil (mg/kg)	32.6
phosphorus	Float	Phosphorus content in soil (mg/kg)	21.3
potassium	Float	Potassium content in soil (mg/kg)	29.0
humidity	Float	Soil humidity percentage	35.7
temperature	Float	Soil temperature in degrees Celsius	26.3
device_id	string (nvarchar)	ID of the ESP32 sensor device that collected the reading	"esp32_001"
created_at	DateTime	Timestamp when this record was created in the database	"2025-01-15 14:30:00"
reading_id	string (nvarchar)	Unique identifier for each shared sensor reading	"shared_001"
is_latest	Boolean (BIT)	Flag indicating if this is the most recent reading (1 = latest, 0 = historical)	1
history_id	string (nvarchar)	Unique identifier for each historical sensor reading	"history_001"

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session_id	string (nvarchar)	Unique identifier for each farmer session	“session_001”
soil_reading_id	string (nvarchar)	Foreign key referencing sensor_data table (links to farmer's copy of reading)	"record_001"
crop_variety	string (nvarchar)	Eggplant variety selected by farmer	“Calixto F1”
crop_stage	string (nvarchar)	Growth stage selected by farmer	“Vegetative”
crop_selected_at	string (nvarchar)	Timestamp when crop selection was made	"2025-01-15 14:35:00"
fertilizer_type	string (nvarchar)	Recommended fertilizer type from ML model	"NPK 14-14-14"
n_amount	Float	Recommended Nitrogen amount in kg/ha	24
p2o5_amount	Float	Recommended Phosphorus (P2O5) amount in kg/ha	56
k2o_amount	Float	Recommended Potassium (K2O) amount in kg/ha	36
fertilizer_kg_ha	Float	Total fertilizer amount recommended in kilograms per hectare	460
application_method	string (nvarchar)	Recommended method for applying fertilizer	“Broadcast”

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feedback_id	string (nvarchar)	Unique identifier for each feedback entry	"feedback_001"
effectiveness	Integer	Rating of recommendation effectiveness (1-5 scale)	4

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notes	string (nvarchar)	Additional notes or comments from the farmer	"Applied slightly less due to weather conditions"
submitted_at	DateTime	Timestamp when feedback was submitted	"2025-01-20 10:15:00"
profile_id	string (nvarchar)	Unique identifier for each farmer profile	"profile_001"
total_sessions	Integer	Total number of sessions recorded for this farmer	13
total_recommendations	Integer	Total number of fertilizer recommendations received	13
average_effectiveness	Float	Average effectiveness rating across all feedback (0.0 to 5.0)	4.1
most_used_variety	string (nvarchar)	Eggplant variety most frequently used by this farmer	"Morado"
most_used_stage	string (nvarchar)	Crop growth stage most frequently selected	"Seedling"
last_updated	DateTime	Timestamp of the last profile update	"2025-01-20 14:30:00"

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Table 4. Data Dictionary

The table above shows the Data Dictionary. There are four columns in the table, first column is the field name of the data. Second column represents to the data types. The third column shows the description of the data, and lastly, the examples or format of the data.

3.3 DEVELOPMENT METHODOLOGY

3.3.1 Process Model

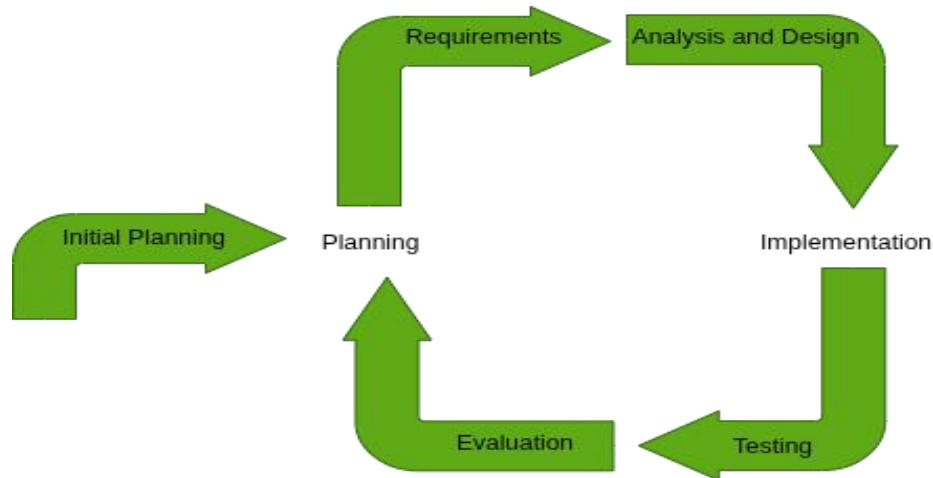


Figure 3.3.1-1. Incremental Model

The construction of the SoilSense is based on an incremental development model, starting with preliminary planning where the project objectives, scope, main elements, and responsibilities are established. Once the first phase, every increment has its own thorough planning procedure, defining particular tasks, resource distribution, and expected deliverables. After planning is done, the project moves into the requirements gathering phase, where nonfunctional and functional needs are determined for that specific increment—e.g., sensor requirements, database requirements, or user interface capabilities.

The analysis and design stage follows, in which the architecture is more defined, data

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flow is tracked, machine learning model structure is defined, and the database schema or mobile app screen layout is designed. Implementation is the final stage in which the highlighted component—sensor integration, machine learning model development, or application construction—has been created and merged with systems like PostgreSQL or cloud services.

Each completed increment is closely tested with unit and integration testing to verify quality and performance standards compliance.

The system then goes through a testing phase in which the performance, the feedback and experiences of the users, and the problems that crop up are evaluated. Experience from this testing is utilized in the next planning cycle. The incremental cycle —planning, building, testing, and evaluating — is iterated through each increment until the system as a whole becomes complete, stable and fully operational. The model gives flexibility, detection of early issues, and capacity to support new discovery or new users' needs throughout the development cycle.

3.3.2 Development Tools

Category	Tool	Description
Programming Language	Kotlin	Primary language for developing the Android mobile application
	Python	Used for training machine learning models and data preprocessing
Database	Firebase Realtime Database	Cloud database for storing sensor data, farmer sessions, and real-time synchronization

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IDE	Android Studio	Integrated development environment for building and testing the Android application
	Arduino IDE	Used for programming and configuring the ESP32 microcontroller
	Visual Studio Code	Used for model development.
Build Tool	Gradle	Dependency management and project building system for Android development
Framework	TensorFlow	Deep learning framework for building and training the fertilizer prediction neural network model
	TensorFlow Lite	Lightweight machine learning framework for on-device inference in Android application
Backend Framework	Firebase SDK (Android)	Software development kit for connecting the Android app to Firebase Realtime Database
Machine Learning Libraries	pandas	Python library for data manipulation and preprocessing of training datasets
	numpy	Python library for numerical operations and array manipulation in model training

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	scikit-learn	Machine learning library providing LabelEncoder, StandardScaler, and train_test_split for data preprocessing
Android Libraries	Jetpack Compose	Modern Android UI toolkit for building native user interfaces
	Android Views	Traditional Android UI components (used alongside Compose)
	Material Design Components	Google's Material Design UI components for Android
	MPAndroidChart	Third-party library for creating charts and graphs in the analytics module
	Retrofit	Type-safe HTTP client for Android, used for API communication
	Firebase Realtime Database SDK	Android SDK for real-time data synchronization with Firebase
Data Format	CSV Files	Used as datasets for training ML models
Microcontroller	ESP32	Microcontroller board for reading soil sensor data via RS485 Modbus protocol
	Arduino Framework	Programming framework for ESP32 microcontroller development

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Table 5. Development Tools

The table above shows the Development Tools. There are three columns in the table, first column is the category of the tools. Second column represents the type of tools. And lastly, the description of development tools.

3.3 TEST METHODOLOGY/PROCEDURES

This section contains the test procedure and methods, the approach taken in making the SoilSense System efficient with an IoT monitoring system from the design stage to deployment.

To ensure that the hardware and software elements work well together, integration testing will take place. Whereas the System testing is done to examine the complete functioning. How easy the system is to use is measured against the user interface. Also, components are tested again to confirm their functioning. Data sensor reports are prepared to tabulate and analyze the results. The results of these testing processes are to analyze and confirm the functioning and usability of the soil sensor system with IoT monitoring to yield reliable data and efficacy of the entire system.

3.4 SYSTEM REQUIREMENTS

HARDWARE

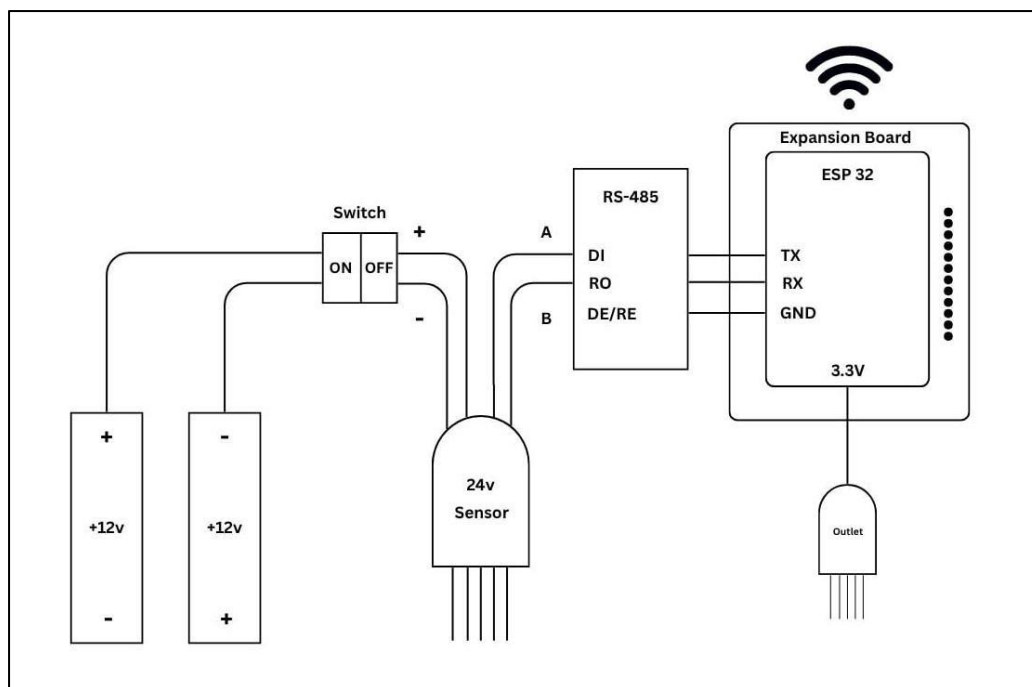
The system being designed consists of essential hardware and software components to enable stable performance in real-time soil monitoring and decision-making. As far as hardware is concerned, the system utilizes the ESP32 Development Board, which is a

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microcontroller that has high processing power as well as a built-in Wi-Fi and Bluetooth capability. It serves as the central controller, which receives and sends data. A Soil Comprehensive Sensor is used to track some of the soil properties such as humidity, temperature, pH, and nutrient content such as nitrogen, phosphorus, and potassium (NPK). The sensor provides real-time data to the ESP32 to be processed. A user device, i.e., a smartphone or tablet, is used as an interface where the user can view and interact with the system using a mobile application.



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Figure 3.4.1 Hardware Wiring Diagram

The figure shows the connection of a 24V soil sensor, an RS-485 communication module, two 12V batteries, and an ESP32 controller to create a complete monitoring system. The two 12V batteries power the soil sensor. The positive line goes through a switch, allowing the user to turn the sensor on or off. The 24V output from the switch feeds directly into the soil sensor. The negative terminal of the battery connects to both the sensor and the RS-485 module, providing a common ground. The sensor sends its readings using RS-485 differential signals through its A and B communication lines. These lines connect to the corresponding A and B terminals on the RS-485 to TTL converter. The converter sends the digital data to the ESP32 through three logic level connections: the DI pin receives data from the ESP32 TX pin, the RO pin sends data to the ESP32 RX pin, and the DE/RE pin connects to an ESP32 GPIO pin to manage transmit/receive mode. The ESP32 gets power separately from a dedicated outlet-powered adapter, ensuring clean and safe voltage levels that are isolated from the 24V sensor supply. Overall, the diagram shows clear separation of power and communication paths, proper grounding, and a safe way to integrate a high-voltage soil sensor with a low-voltage microcontroller.

HARDWARE COMPONENTS

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CATEGORY	DEVICE NAME	IMAGE	AMOUNT
Microcontroller	ESP32 Dev Module		₱ 349
	ESP32 Expansion Board		₱ 119
Sensors	RS485 Output 5-Pin Soil Sensor		₱ 3,000
	RS485 Signal Converter		₱ 31
Prototyping & Wiring	Breadboard		₱ 108
	Jumper Wires		₱ 194
	Data Cable		₱ 189
Power Supply	Battery 12Volts		690 x 2 ₱1,380

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	Switch (Single Gang)		₱ 35
	Power Adaptor		₱ 91
Internet Connection	Pocket Wifi		₱508
Hardware Case	(plywood,Screws,Nails, Paint,Brush)		₱ 350

SOFTWARE

At the software level, the system microcontroller is written in the Arduino IDE, enabling communication between the ESP32 and the sensors. The smartphone application is created using

Android Studio with the aid of Kotlin and Java — Android - oriented programming languages.

The backend of the system runs on Firebase, a cloud -oriented system utilized to store the sensor readings and handle the authentication of users and facilitate the provision of the real - time update. The hardware and software combination collectively guarantees the provision of intelligent data -driven information to aid sustainable and efficient agricultural production.

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3.5 DATASET METHODOLOGY

This section presents the methodology used to construct the dataset employed in training the fertilizer recommendation model. The dataset was synthetically generated but grounded in scientifically validated agronomic principles, fertilizer application guidelines, and soil nutrient sufficiency standards. To ensure that the machine learning model receives realistic, agriculturally sound, and Philippine-contextual input, the dataset integrates documented nutrient ranges, environmental tolerances, crop stage requirements, and fertilizer response logic obtained from reputable agricultural agencies and institutions.

The methodology draws from multiple authoritative references, including the **Department of Agriculture – Bureau of Soils and Water Management (DA-BSWM, 2022)**, the **DA–High Value Crops Development Program (HVCDP) Eggplant Production Guide**, the **UPLB Institute of Plant Breeding (2021)** varietal descriptions, the **Food and Agriculture Organization (FAO, 2020)** soil fertility guidelines for Solanaceae crops, and the **DA–BAR Fertilizer and Nutrient Management Manual (2023)**. These sources informed the nutrient thresholds, crop stage needs, fertilizer types, and soil type behavior reflected in the dataset.

3.5.1 Dataset Overview

The dataset, saved as *phil_eggplant_v2_balanced.csv*, consists of more than 7,700 rows of synthesized eggplant field conditions. Each row represents a scenario that combines sensor measurements, crop stage, soil type, eggplant variety, and the corresponding fertilizer recommendation generated through a rule-based agricultural logic system. Although the

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dataset is computationally generated, every value and fertilizer decision is anchored on real agronomic standards, ensuring realism and scientific validity.

3.5.2 Dataset Variables

The dataset contains fourteen (14) variables categorized into soil sensor attributes, contextual field descriptors, fertilizer outputs, and nutrient requirement indicators. Soil sensor variables include nitrogen (mg/kg), phosphorus (mg/kg), potassium (mg/kg), pH, moisture (%), and temperature (°C). These represent measurable indicators obtained through soil sensors or laboratory analysis.

Contextual field variables include crop growth stage, which corresponds to seedling, vegetative, flowering, or fruiting. Eleven widely cultivated Philippine eggplant varieties were incorporated to reflect real varietal diversity, including Dumaguete Long Purple, Casino, Purple King, Davao Beauty, and Calixto F1, based on UPLB and DA varietal listings. The dataset also includes fertilizer output variables, particularly the recommended fertilizer material and application method. Lastly, the values for recommended N, P_2O_5 , and K_2O (kg/ha) per row represent stage-dependent nutrient demand calculations derived from DA and FAO guidelines.

3.5.3 Derivation of Soil Nutrient Ranges

The nutrient values in the dataset were generated by referencing documented sufficiency thresholds from BSWM soil test interpretation tables and FAO standards for Solanaceae crops. Nutrient ranges vary by crop stage to reflect changing nutrient uptake patterns. During

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the seedling stage, nutrient demand is low; hence nitrogen values ranged between 15–35 mg/kg, phosphorus between 10–25 mg/kg, and potassium between 15–30 mg/kg. The vegetative stage exhibits the highest nitrogen requirement, with values ranging from 25–55 mg/kg. Phosphorus and potassium values during this stage reflect moderate demand, in alignment with DA-BAR fertilization recommendations.

During flowering, the phosphorus requirement increases as reproductive structures develop; thus phosphorus values were generated within the 12–30 mg/kg range. Potassium values were also elevated during fruiting (25–60 mg/kg), following the known role of K in fruit enlargement and quality. These stage-specific nutrient ranges ensure that the dataset captures biologically consistent soil fertility behavior.

3.5.4 Fertilizer Recommendation Ground Truth Assignment

The fertilizer recommendation for each dataset row was produced using a carefully designed rule-based logic grounded in established agricultural guidelines. When nitrogen levels fell below 25 mg/kg, the model assigned Urea (46-0-0), following DA recommendations for nitrogen deficiency correction. When phosphorus levels dropped below 15 mg/kg, the dataset assigned DAP (18-46-0), the standard phosphorus-rich fertilizer recommended by BSWM and

FAO. Potassium deficiency below 20 mg/kg triggered the assignment of Muriate of Potash (00-60), consistent with potassium supplementation guidelines.

In cases where nutrient levels fell within established sufficiency zones, the recommended fertilizer was NPK 14-14-14, identified in the DA-HVCDP guide as the standard balanced fertilizer for general eggplant nutrition. When nutrient levels were consistently high across

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all major elements, the assigned recommendation shifted to Vermicast or organic compost, reflecting DA-BAR guidelines promoting organic amendments when soil fertility is already elevated. These rules formed the ground truth labels used to train the machine learning model.

3.5.5 Dataset Balancing Procedures

To prevent model bias and ensure that all growth stages, soil types, and eggplant varieties were equally represented, the dataset was generated using balanced distribution mechanisms. Each of the eleven varieties was assigned approximately 700 rows, resulting in more than 7,700 total instances. Crop growth stages were evenly distributed across all varieties. This balancing ensures the model's ability to generalize across different agronomic conditions and reduces the likelihood of prediction bias.

3.6 Sensor Calibration and Validation

3.6.1 Soil Testing Procedure

To evaluate the performance of the developed soil sensor, soil testing was conducted using a representative soil sample. The sensor was initially tested in air to establish baseline readings, after which it was inserted into the soil sample. Gradual amounts of water were added to the soil to observe variations in moisture content, pH level, and nutrient readings.

Following the initial testing, the sensor underwent calibration using laboratory soil analysis results as reference values. The laboratory analysis was conducted by the

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Regional Soil Laboratory in Lipa City, Batangas, which served as the ground truth for validating the accuracy of the sensor. After calibration, the sensor was re-tested on the same soil sample, and the calibrated readings were recorded for comparison.

This process ensured that the sensor readings before and after calibration could be systematically evaluated against laboratory-standard measurements.

3.6.2 Comparison of Soil Test Results

Table 3.6.2.1 presents the comparison between the uncalibrated sensor readings, calibrated sensor readings, and laboratory soil test results. The comparison focuses on soil pH, nitrogen (N), phosphorus (P), and potassium (K) concentrations.

Parameter	Soil Laboratory Result	Sensor (Before Calibration)	Sensor (After Calibration)
pH	5.78	6.48	6.14
Nitrogen (ppm)	20.16	22	33
Phosphorus (ppm)	18.48	31	18.69
Potassium (ppm)	163.01	119	152.61

Table 3.6.2.1. Comparison of Soil Test Results

Before calibration, noticeable deviations were observed between the sensor readings and laboratory results, particularly in phosphorus and potassium values. After calibration, the



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sensor readings demonstrated closer agreement with the laboratory reference values, indicating improved accuracy of the system.

3.6.3 Error Computation

To quantitatively assess the accuracy of the calibrated sensor, error values were computed by comparing the calibrated sensor readings with the laboratory soil test results.

Error Formula

$$\text{Error} = \text{Sensor Value} - \text{Laboratory Value}$$

The laboratory soil test values were considered the reference or true values due to their standardized analytical procedures. The sensor values after calibration represent the system's measured output. The difference between these values indicates the deviation of the sensor from the reference measurement.

Individual Error Computation

Parameter	Individual Error Computation
pH	$6.14 - 5.78 = 0.36$
Nitrogen (ppm)	$33.00 - 20.16 = 12.84$
Phosphorus (ppm)	$18.69 - 18.48 = 0.21$
Potassium (ppm)	$152.61 - 163.01 = -10.40$

Table 3.6.3.1. Individual Error Computation

Parameter	Laboratory Value	Sensor Value	Error
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		(Calibrated)	
pH	5.78	6.14	+0.36
Nitrogen (ppm)	20.16	33	+12.84
Phosphorus (ppm)	18.48	18.69	+0.21
Potassium (ppm)	163.01	152.61	-10.40

Table 3.6.3.2. Individual Error Values

3.6.4 Mean Absolute Error (MAE) Calculation

To further evaluate the overall accuracy of the calibrated sensor, the Mean Absolute Error (MAE) was computed for nitrogen, phosphorus, and potassium. pH was excluded from the MAE computation because it is unitless and not expressed in parts per million (ppm), unlike N, P, and K.

MAE Formula

$$MAE = \frac{1}{n} \sum |Error|$$

MAE Computation

$$MAE = \frac{|12.84| + |0.21| + |-10.40|}{3}$$

$$MAE = \frac{23.45}{3}$$

$$MAE = 0.88$$

The results indicate that sensor calibration significantly improved the agreement between

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sensor readings and laboratory soil test values. Among the nutrients measured, phosphorus exhibited the lowest error, suggesting high accuracy in its detection after calibration. Nitrogen showed the largest deviation, which may be attributed to variations in soil composition, sensor sensitivity, or environmental factors affecting nutrient measurement.

Despite these variations, the overall Mean Absolute Error (MAE) of 0.88 demonstrates a substantial improvement in sensor accuracy and indicates that the calibrated system is capable of providing reliable soil nutrient measurements. The findings suggest that the developed soil sensor can serve as an effective tool for preliminary soil analysis and decision support in agricultural applications.

3.6.5 Limitations of the Calibration and Validation Process

This study focused on validating the accuracy of the developed soil sensor using a single representative soil sample and laboratory reference data. While the results demonstrate improved accuracy after calibration, variations in soil type, moisture level, and environmental conditions were not extensively explored. Future studies may include multiple soil samples from different locations and repeated trials to further assess the consistency and robustness of the sensor across varying conditions.

3.7 QUALITY PLAN

The system's quality will be evaluated using the ISO/IEC 25010:2011 standard, which defines a comprehensive model for assessing software product quality. This framework focuses on eight key characteristics: **functional suitability, performance efficiency,**

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compatibility, usability, reliability, security, maintainability, and portability. These characteristics ensure that the system meets both technical and user-centered requirements.

To conduct a well-rounded quality assessment, the researchers will employ targeted procedures and tools that align with each ISO 25010 quality attribute. For instance, functionality and usability will be assessed through user surveys and testing scenarios, while reliability and performance efficiency will be measured by monitoring system responsiveness and fault tolerance under varying conditions.

A structured questionnaire form will be a main assessment tool, gathering user responses to see how successfully the system addresses real -world agricultural requirements. That information will be used to make improvements and confirm that the system meets international standards, so it is precise, user -friendly, reliable, and productive for the purpose it is designed for.

3.8 EVALUATION PLAN

The system validation of the SoilSense systems will comply with the ISO/IEC 25010 standard in order to guarantee that the system is adequate with respect to the specified software quality attributes. These foci of attention are functional suitability, reliability, usability, performance efficiency, security, maintainability, and portability.

System quality will be exercised through controlled test processes that will compare the validity of the soil sensor readings, the precision of machine learning forecasts, and the convenience of the mobile app. Evaluation instruments will consist of a Likert- s cale

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questionnaire survey, to determine the level of user satisfaction and usability and system efficiency indicators like model precision and data latency of responses. The evaluation will guarantee the system performs well in actual world agricultural applications in providing efficient and convenient crop and fertilizer advice.

The gathered data were tallied, analyzed, and subjected to statistical treatment. Responses were based on a scale with the following values:

Likert Scale for 5-point scale:

Range	Descriptive Rating	Verbal Interpretation
4.21 – 5.00	Highly Acceptable	Strongly Agree
3.41 – 4.20	Acceptable	Agree
2.61 – 3.40	Moderately Acceptable	Neutral
1.81 – 2.60	Less Acceptable	Disagree
1.00 – 1.80	Not Acceptable	Strongly Disagree

Percentage Formula:

$$p = \frac{f \times 100}{n}$$

P = Percentage

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f = frequency

n = total number of respondents

Weighted Mean:

$$\bar{x} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

$\bar{X}$ = Weighted Mean

W_i = weight associated with each value X_i

X_i = individual data points

N = total number of data points f_i

f_i = frequency of the data point X_i

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CHAPTER 4 – RESULTS AND DISCUSSION

Data Interpretation Techniques

The data gathered from the SoilSense system and the responses from the user evaluation surveys were interpreted using clear and organized procedures to understand how well the system performed. The readings collected by the sensors such as NPK levels, pH, moisture, temperature, and humidity—were compared with standard soil ranges from agricultural references. By this, the researchers could determine whether the system had correctly identified the soil conditions, and the fertilizer recommendation generated from the model was correct for the detected soil state. The feedback provided through the survey forms, regarding user evaluation, was interpreted by analyzing how the respondents rated the system according to ISO 25010 criteria. Weighted means and descriptive ratings supported the conversion of numerical scores into clear interpretations that reflected the experiences of the users. Graphs and tables have been included to present the results in visual format for easier comprehension of trends, level, and perception of the SoilSense system. These methods of interpretation provided the researchers with organized and meaningful under-standings of both the technical outputs and the users' evaluations.

Data Analysis Techniques

A quantitative approach was employed to quantify user feedback on the SoilSense system using ISO 25010 criteria. Weighted means and descriptive equivalents were computed to summarize overall user satisfaction and to identify which specific quality attributes met expectations or required further refinement. For the technical outputs, descriptive statistics

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were used to assess the stability of repeated sensor readings, alongside direct comparisons with laboratory soil test results and field observations, including ongoing comparisons between plots using SoilSense recommendations and those following traditional practices. These analytic procedures ensured that both quantitative evidence and operational performance indicators were captured objectively, providing a reliable basis for the discussion of SoilSense outcomes in this chapter.

Part I. Demographic Profile of the Respondents

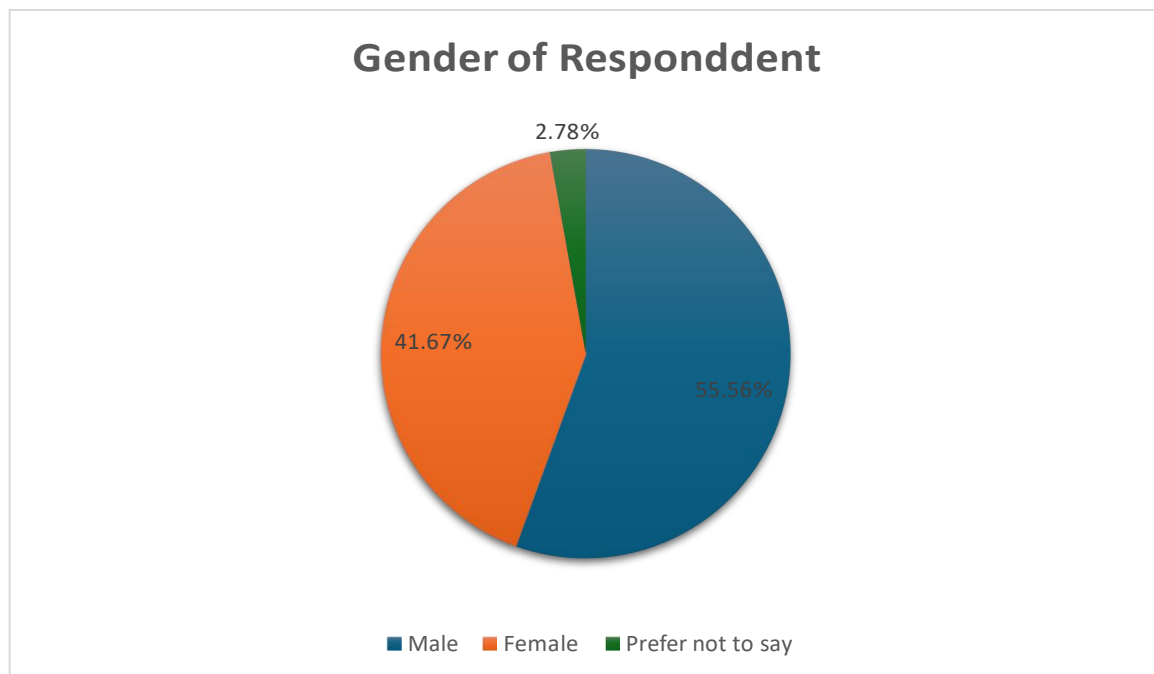


Figure 4.1 Gender of Respondent

Figure 4.1 shows the gender of the respondents. Out of the 36 respondents surveyed, 55.56% (20) are males, and 41.67% (15) are females, while 2.78% (1)



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prefer not to say their gender, which means majority of the total sample population are male farmers.

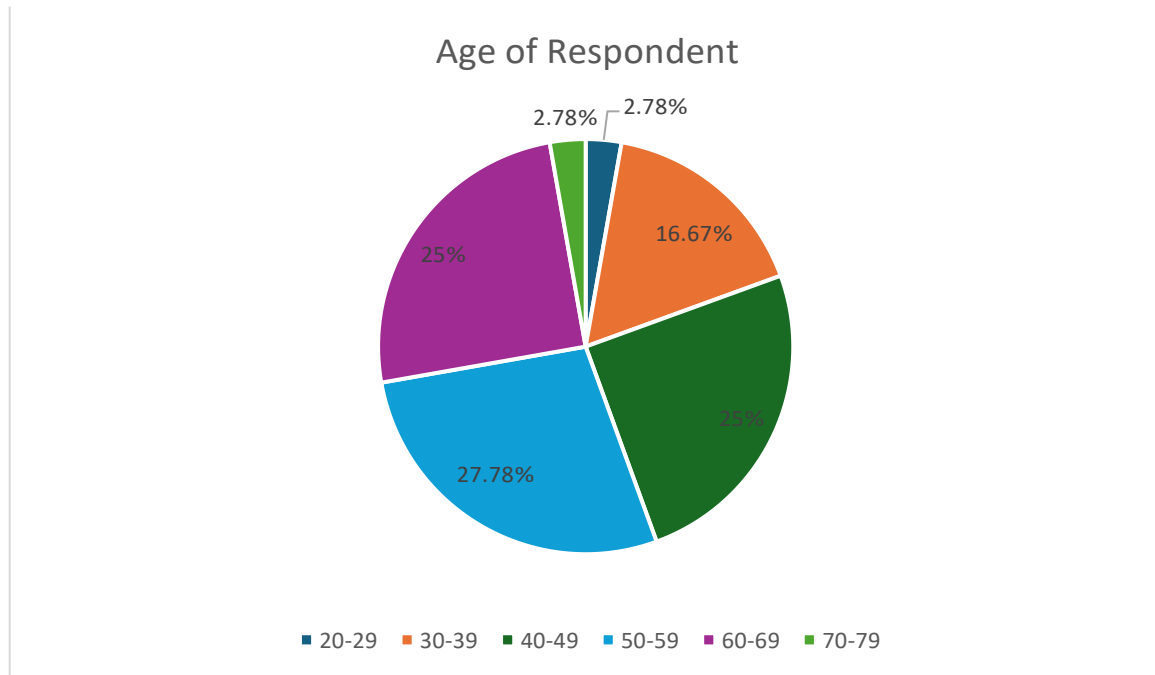


Figure 4.2 Age of Respondent

Figure 4.2 shows the age range of the respondents. The majority (27.78%) of the respondents belong to the 50-59 age range. The age 40-49 and 60-69 comes in second, which comprises 25% of the total respondents. The lowest percentage are 20-29 and 70-79 years old, which only comprises 2.78% of the total sample population.



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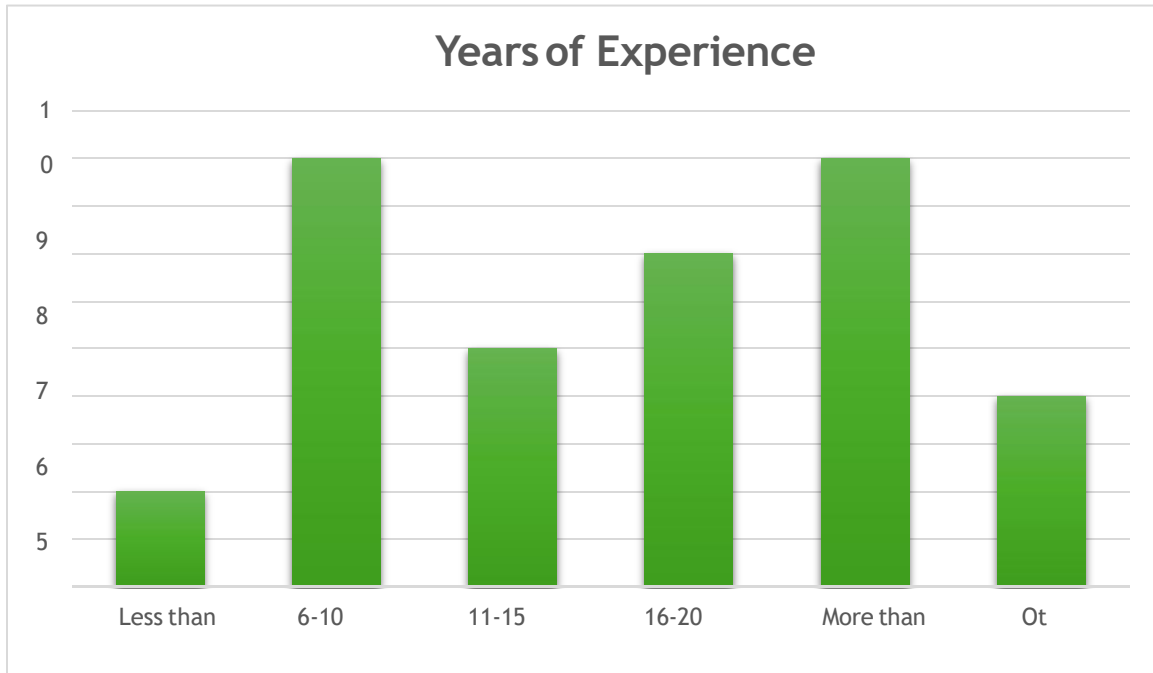


Figure 4.3 Years of Experience as Farmers

Figure 4.3 presents the frequency and percentage distribution of the respondents' years of experience as farmers. The figure shows that the majority of the respondents have 6–10 years of farming experience or more than 21 years of experience, while the minority are those with less than 5 years of experience.



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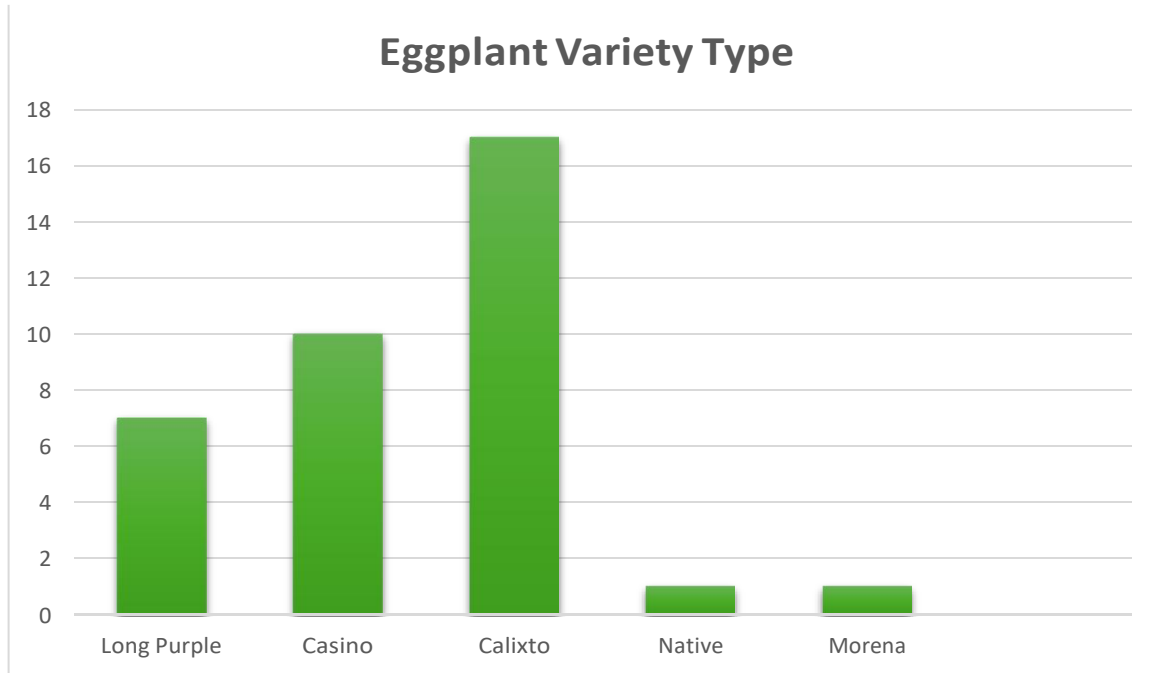


Figure 4.4 Eggplant Variety

Figure 4.4 presents the frequency and percentage distribution of the respondents' eggplant variety type. The figure shows that the majority (17) of the respondents have a variety of Calixto, while the minority (1) are those who have Native and Morena variety.



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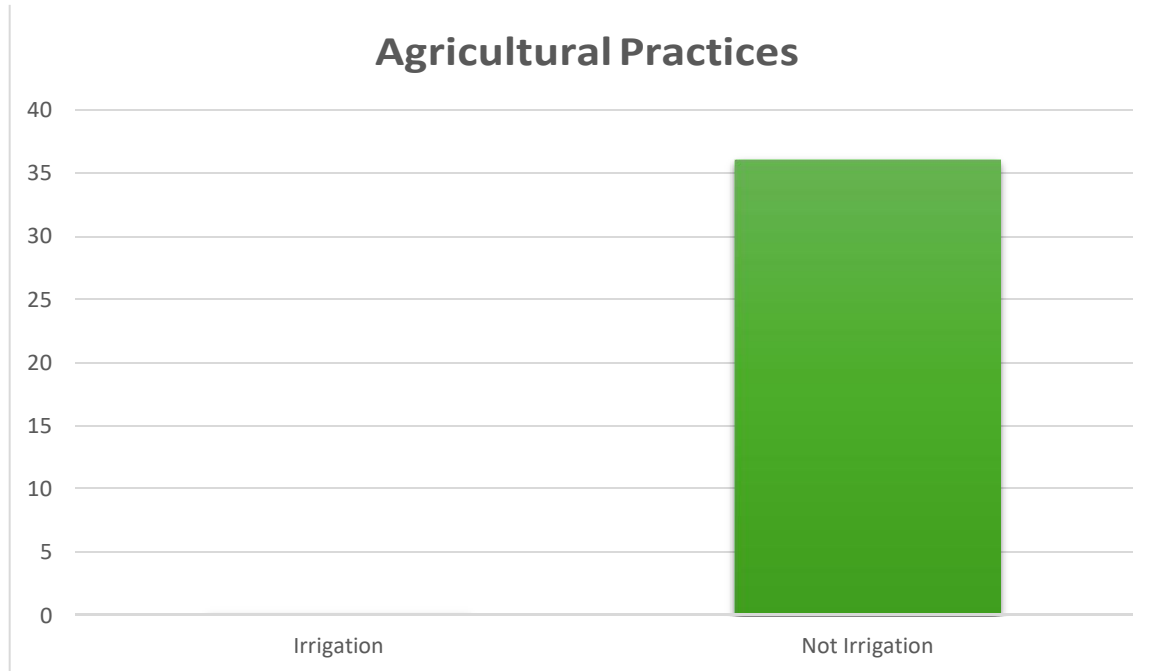


Figure 4.5 Agricultural Practices

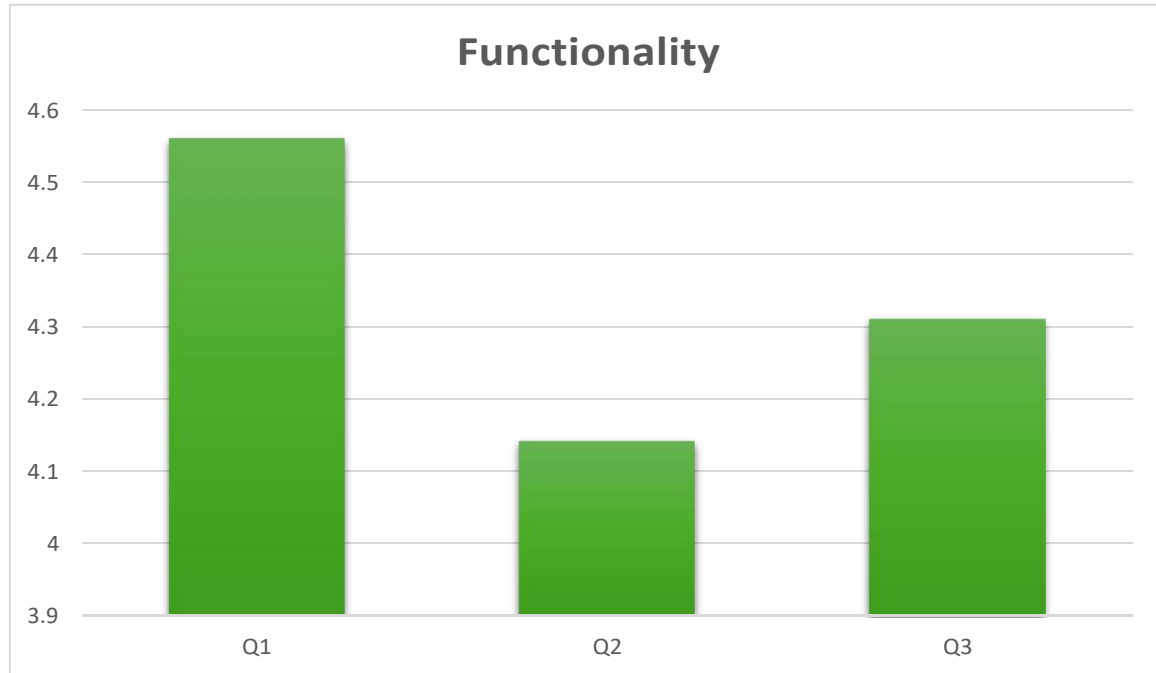
Figure 4.5 presents the frequency and percentage distribution of the respondents' agricultural practices. The figure shows that all (36) of the respondents do not use irrigation practices for their farming.

Part II: System Quality Evaluation (ISO 25010)

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*Figure 4.6 Summary of the Evaluation Outcomes Regarding
SoilSense Functionality*

Figure 4.6 illustrates the evaluation results for SoilSense functionality, providing an overview of its efficacy, highlighting strengths, and identifying areas for improvement. For the functional suitability category, the highest rating of 4.56 (Strongly Agree) was given to the system's capability to accurately measure soil pH, nitrogen, phosphorus, potassium, temperature, and humidity. This demonstrates the system's effectiveness in classification the soil health nutrients, which is one of its primary objectives.

Conversely, the lowest rating of 4.14 (Agree) was given to crop and fertilizer recommendations being relevant to local conditions. While this score still falls within the acceptable range, it indicates a need for improvement in recommendation related to the local conditions.



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Despite this minor shortcoming, the overall functionality score of SoilSense falls within the highly acceptable range, confirming its dependability and ability to meet user expectations. The consistent score across the evaluation reflects a stable and reliable system that fulfills its functional requirements.

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Figure 4.7 Summary of the Evaluation Outcomes Regarding SoilSense Performance

Figure 4.7 presents the evaluation results for SoilSense performance, offering a complete assessment of its effectiveness, emphasizing its strengths and pinpointing areas for improvement.

For the performance efficiency category, the highest rating of 4.19 (Agree) was given to the system processing the data quickly and provides timely recommendations. This demonstrates the system's ability to operate efficiently under normal conditions, meeting user expectations for real time analysis and responsiveness.

On the other hand, a rating of 4.06 (Agree) was given to the sensor readings updating in near real time without noticeable delay, the same rating (4.06) was given to the mobile app



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having the capability to load quickly and use minimal mobile data. These findings reaffirm the sensor readings update in near real time without noticeable delay, and the mobile app loads quickly while using minimal mobile data.

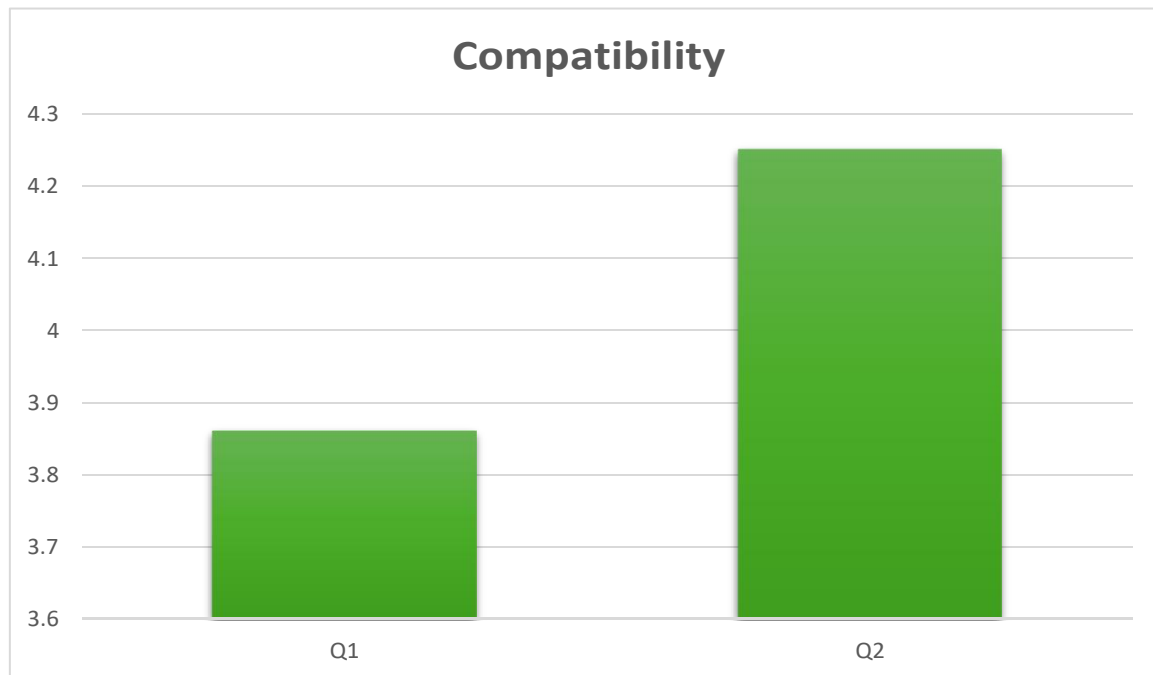


Figure 4.8 Summary of the Evaluation Outcomes

Regarding SoilSense Compatibility

Figure 4.8 illustrates the evaluation results for SoilSense compatibility, providing insights into its strengths and areas for improvement based on two questions.

For the compatibility category, the highest rating of 4.25 (Strongly Agree) was given to SoilSense integrating smoothly with farmer's existing farming practices. This demonstrates that the users view the system integration with current structure as highly acceptable and a notable strength.

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However, the lower rating of 3.86 (Agree) was given to the application's compatibility to work well on the respondents' smartphone model and Android version. These scores fall within the acceptable range and feedback indicates that this area require additional attention to improve user experience and satisfaction.

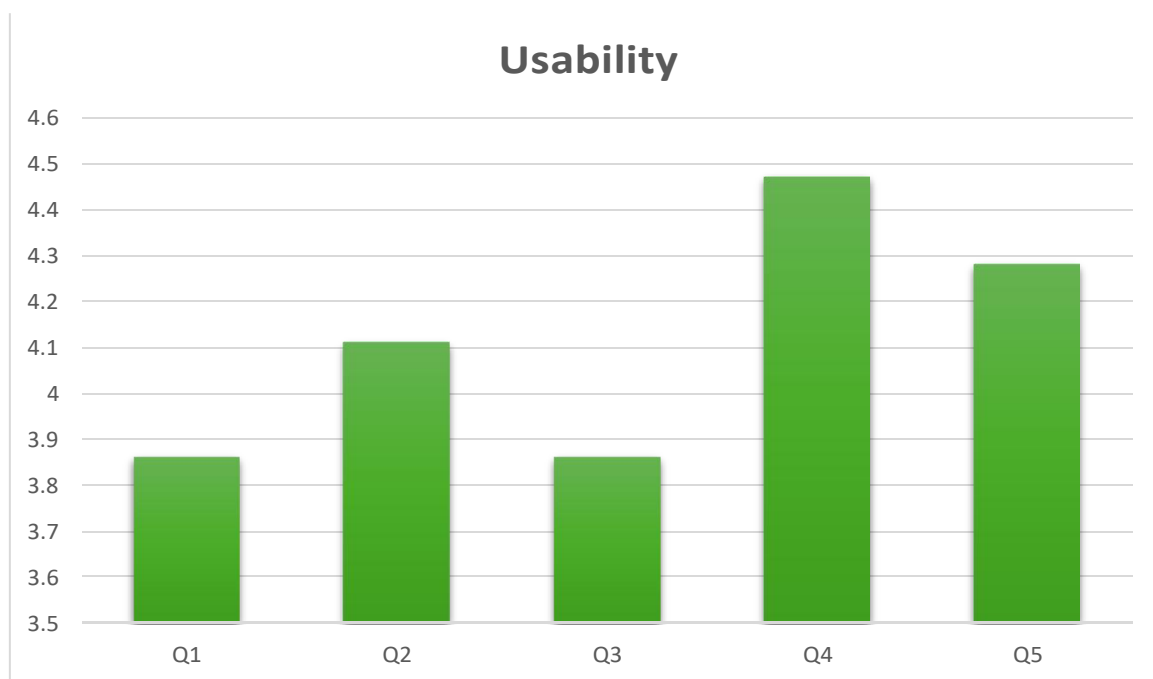


Figure 4.9 Summary of the Evaluation Outcomes

Regarding SoilSense Usability

Figure 4.9 illustrates the evaluation results for SoilSense interaction, assessed based on four questions.

For the interaction capability category, the highest rating of 4.47 (Strongly Agree) was given to the system improving the farming decisions of farmers. This scores a highly favorable



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perception of the system usability and accessibility, with respondents viewing it as highly acceptable.

However, slightly lower ratings of 3.86 (Agree) were given to the app's user interface being clear and easy to navigate. Q3 achieves the same rating (3.86) which pertains to users learning to use the system without outside help. While these scores remain within the "Agree" range, that is closer to the lower end of acceptability, suggesting room for improvement in these areas.

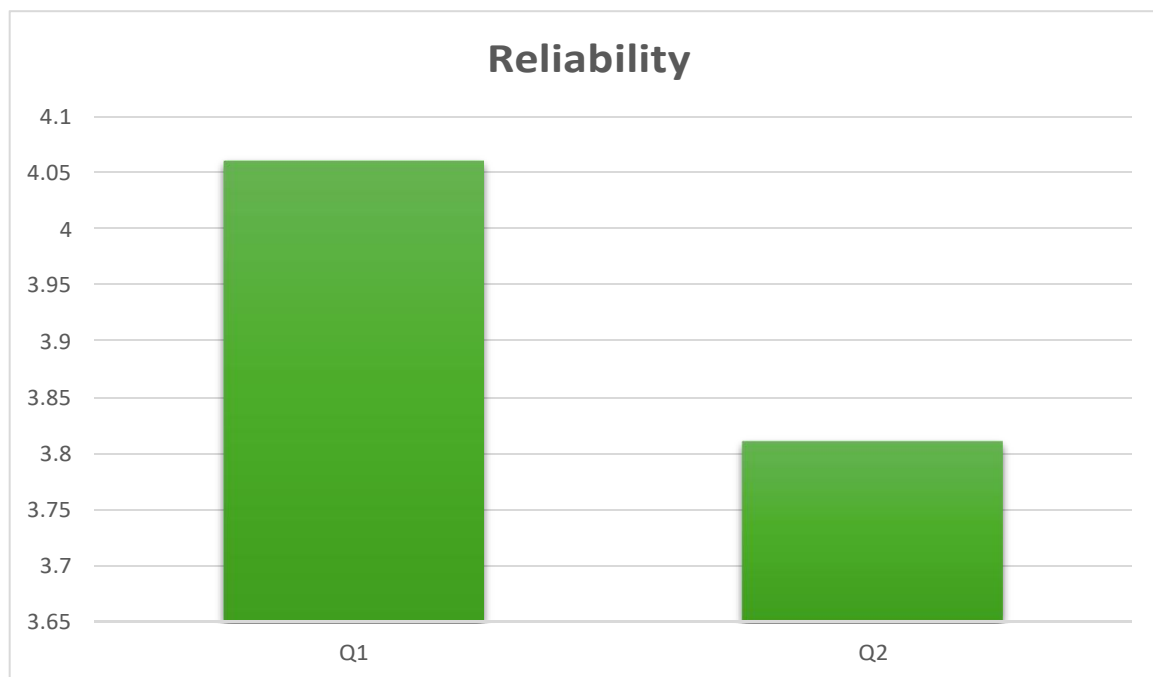


Figure 4.10 Summary of the Evaluation Outcomes

Regarding SoilSense Reliability

Figure 4.10 presents the evaluation results for SoilSense dependability, assessed across two criteria.



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For the reliability category, the highest rating of 4.06 (Agree) given to the sensor data remaining consistent during repeated measurements. This score indicates that respondents generally regard SoilSense reliability as acceptable, reflecting a reasonable level of confidence in their consistent performance.

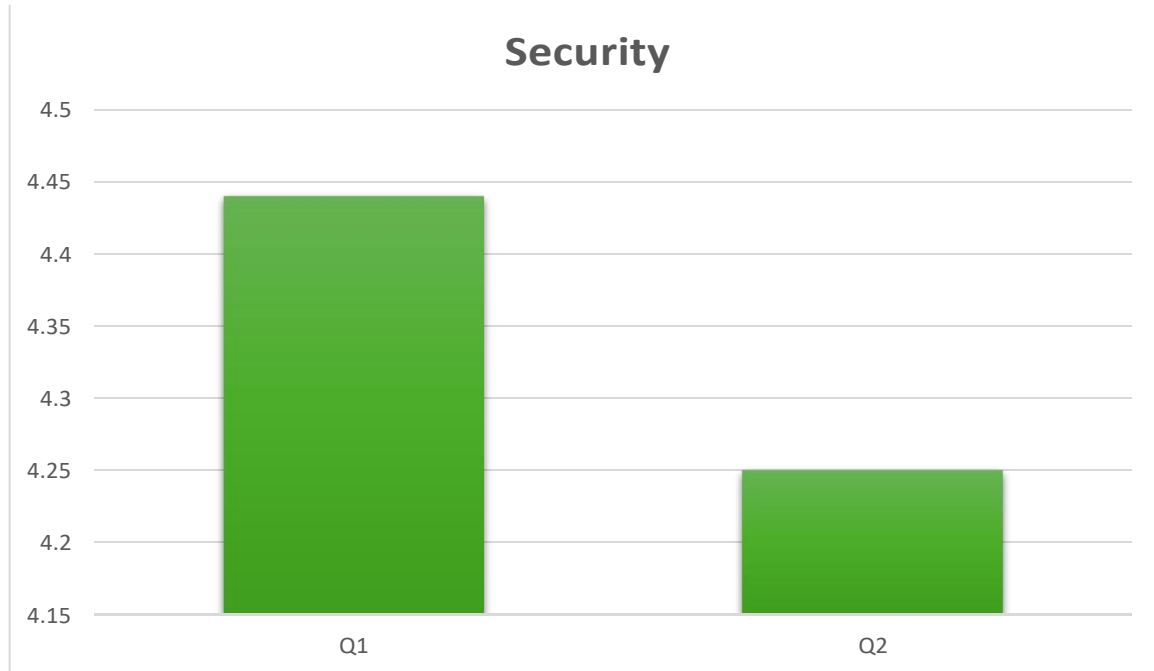
On the other hand, the system capability to function even after short internet outages received a lower score of 3.81 while still within the “Agree” range, this score is on the lower end, highlighting an area where SoilSense dependability could be enhanced to ensure more consistent performance in demanding scenarios.

Overall, the findings confirm that SoilSense dependability is generally recognized as acceptable. However, the lower score in Q2 suggests an opportunity for targeted improvements to strengthen the system’s reliability under high-traffic conditions, ultimately enhancing user satisfaction.

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*Figure 4.11 Summary of the Evaluation Outcomes
Regarding SoilSense Security*

Figure 4.11 illustrates the evaluation results for the SoilSense system's security, assessed using two specific questions. The highest mean score of 4.44 (Strongly Agree) was obtained for Q1, which indicates that users feel confident that their personal and farm data are protected and kept private by the SoilSense system. Meanwhile, a mean score of 4.25 (Strongly Agree) was given to Q2, which reflects the users' agreement that the absence of a login process helps prevent the need to provide sensitive and personal information. Overall, these results suggest that users have strong trust in the system's ability to safeguard their data.



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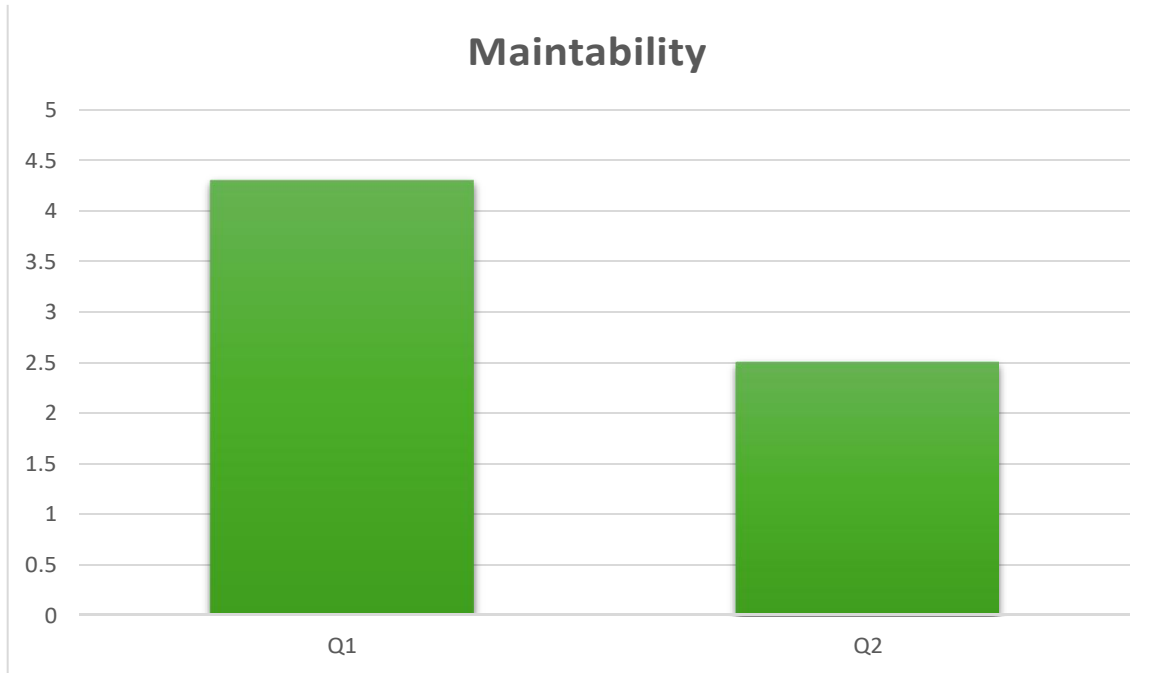


Figure 4.12 Summary of the Evaluation Outcomes

Regarding SoilSense Maintainability

Figure 4.12 illustrates the evaluation results for SoilSense maintainability, assessed based on two questions.

For the maintainability category, the highest rating of 4.25 (Strongly Agree) was given to the system updates or fixes being able to be installed easily. This indicates that users perceive the SoilSense system as adaptable and convenient when it comes to applying updates or improvements.



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Meanwhile, the criterion assessing the device maintenance and its calibration being simple, received a slightly lower score of 4.14 (Agree). While still within the acceptable range, this score suggests that users may encounter minor challenges or complexities in maintaining or calibrating the device.

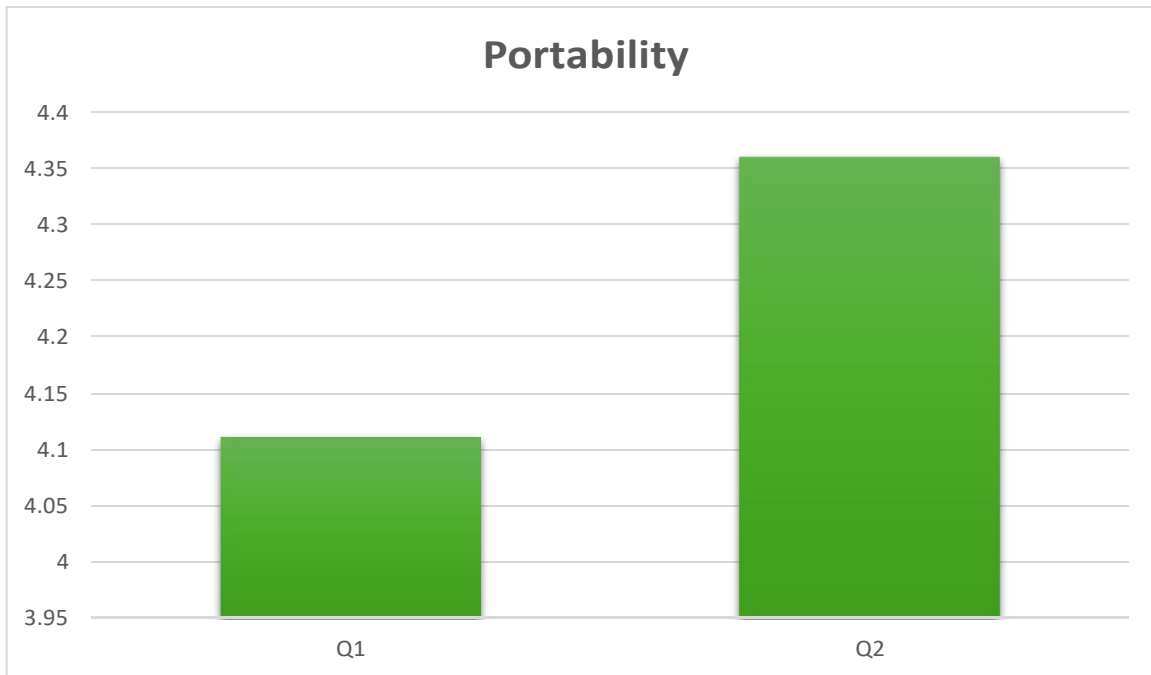


Figure 4.13 Summary of the Evaluation Outcomes

Regarding SoilSense Portability

Figure 4.13 illustrates the evaluation results for SoilSense flexibility, based on two questions. Q2 “*The app can be reinstalled on another phone without issues*” scored 4.36 (Agree). This result reflects a positive perception of SoilSense ability to perform consistently and adapt to various deployment scenarios.



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While the statement “*I can move the sensor to another field and reconnect it easily*” receive a slightly lower score of 4.11(Agree). While still acceptable, this score suggests that users may encounter minor challenges during the installation process. Addressing these challenges could enhance the system flexibility and improve overall use satisfaction.

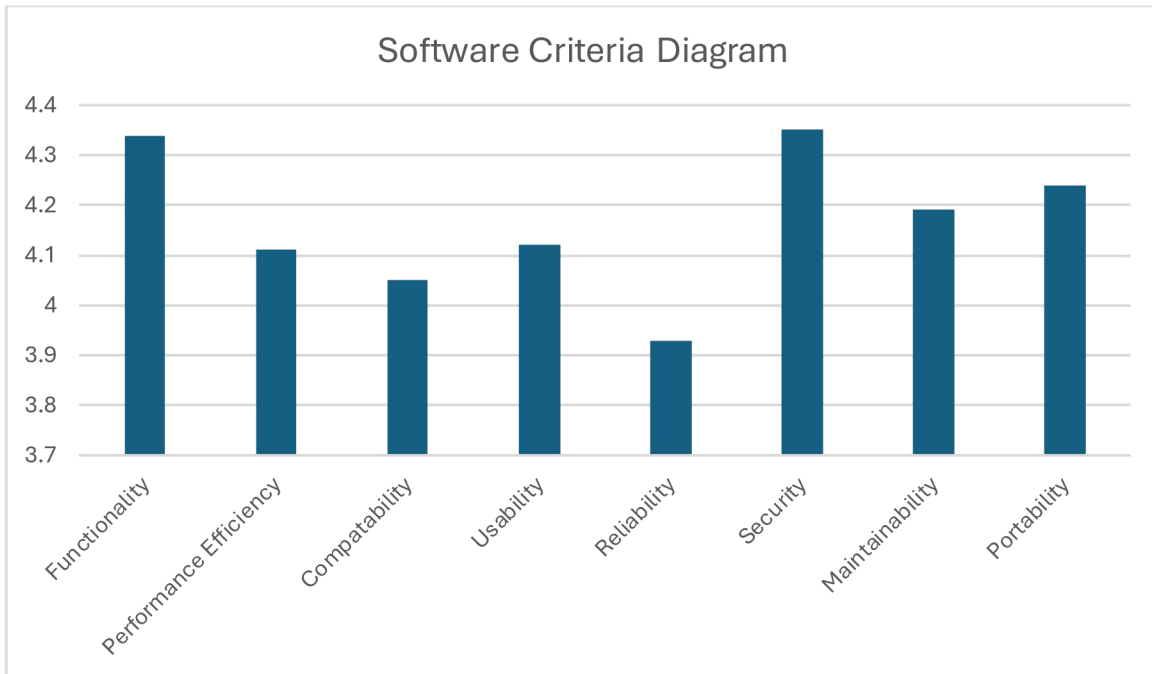


Figure 4.14 Summary of the Evaluation Result of SoilSense

Figure 4.14 presents the evaluation results of SoilSense, illustrating the user feedback on the system’s performance across multiple criteria.

Starting with Security, the system achieved the highest score of 4.35 (Strongly Agree), reflecting strong performance in its ability to protect sensitive data and ensure secure



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operation. This underscores the effectiveness of features such as protection against unauthorized access, which are crucial for maintaining user trust and securing information.

Functionality received a rating of 4.33 (Strongly Agree), reflecting users' confidence in the SoilSense system's ability to deliver accurate soil readings and operate reliably under varying field conditions.

Portability received an acceptable rating of 4.24 (Strongly Agree), indicating that users found the SoilSense system reasonably easy to transport, set up, and utilize across different field locations.

Maintainability received a rating of 4.20 (Agree), suggesting that users found the SoilSense system manageable to service, update, and troubleshoot when necessary.

Usability (4.12, Agree) received acceptable ratings, indicating areas with potential for further optimization. While the system improvements in these aspects could enhance overall usability and ensure smoother long-term update and maintenance.

In contrast, Performance (4.10, Agree) were identified as critical areas for improvement. The Performance score reflects concerns about responsiveness, particularly under high workloads or traffic conditions. Addressing these issues is vital for improving operational efficiency and ideal integration.

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Compatibility (4.06, Agree) the compatibility ratings indicate challenges in integrating SoilSense with existing traffic management platforms and various devices.

Lastly, Reliability, with a score of 3.94 (Agree), highlight some inconsistencies in system performance that could impact user confidence. These fluctuations, particularly during peak usage, underline the need for targeted improvements to ensure consistent operation.

In summary, the evaluation results for SoilSense reveal its strengths in Security, Functionality, and Portability, which provide a solid foundation for its effectiveness. The overall mean rating of **4.17**, which falls within the “Agree” range, indicates that users have a generally positive perception of the SoilSense system across all ISO 25010 quality criteria. This score reflects the system’s satisfactory performance in terms of accuracy, usability, portability, security, and maintainability, while also highlighting areas that may require further refinement such as compatibility, reliability, and performance efficiency. Overall, the mean score demonstrates that SoilSense successfully meets user expectations and provides meaningful value in supporting soil assessment and decision-making for farmers.

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Growth Timeline of Eggplant from Seed to Fruiting Stage

Growth Stage	Description	Approximate Duration
Seed Germination	Emergence of seedlings from seeds under favorable temperature and moisture conditions	6–14 days after sowing
Seedling Stage	Development of primary leaves, stem strengthening, and root establishment	4–8 weeks after germination
Transplanting Stage	Transfer of seedlings from nursery to prepared soil plots	~30–45 days after sowing
Vegetative Growth	Rapid increase in plant height, leaf expansion, and stem development	2–4 weeks after transplanting
Flowering Stage	Emergence of flowers indicating the start of reproductive growth	~50–60 days after sowing
Fruiting Stage	Fruit initiation and enlargement following successful pollination	7–14 days after flowering
Harvest Maturity	Fruits reach marketable size and quality	~100–120 days after sowing or 65–85 days after transplanting

Table 4.1 Growth Timeline of Eggplant from Seed to Fruiting Stage

This table presents the standard growth stages of eggplant based on established agricultural references and serves as a basis for comparing expected and actual growth observed in the study.



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Seed Germination and Early Seedling Development

The germination and early seedling growth of the *Calixto F1*, *Long Purple*, and *Casino* eggplant varieties were monitored from sowing until the fourth week of development. The observations documented the transition from seed germination to the early seedling stage, to determine the readiness of the plants for transplanting.



Day 0 – November 4, 2025 (Sowing)

The seeds of *Calixto F1*, *Long Purple*, and *Casino* eggplants were sown in a seedling tray. Seed trays were used because eggplant seeds are delicate and should not be planted directly in soil, ensuring better germination and early growth.



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Day 9 – November 13, 2025 (Early Germination Progress)

Nine days after sowing, several seeds showed early signs of germination. While most seeds had not yet emerged above the soil surface, a few seeds displayed slight cracking and swelling, indicating that the sprouting process had begun. These observations suggested that germination was in progress, although full sprouting had not yet occurred for the majority.



WEEK 3 – November 21, 2025 (Germination to Seedling Transition)

By the eighteenth day after sowing, the seeds had successfully germinated. Visible stems and cotyledon leaves were present in multiple seedlings across all three varieties. This stage

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marked a clear transition from germination to the seedling stage. The development of the initial leaves indicated that the seedlings had begun photosynthetic activity necessary for subsequent growth.



Week 4 – November 25, 2025 (Seedling Growth Stage)

During the fourth week, seedlings showed noticeable improvement in stem strength and leaf development. Several seedlings had produced additional true leaves, indicating healthy vegetative progress. At this point, the seedlings were approaching readiness for transplanting, as they already exhibited established stems and an expanded leaf canopy.

Evaluation of Seed Germination and Seedling Development Assisted by SoilSense The germination and early seedling development of *Calixto F1*, *Long Purple*, and *Casino* eggplants were observed over a four-week period to evaluate initial growth performance in relation to the soil conditions analyzed through the SoilSense application. Prior to sowing, soil samples from the planting medium were subjected to laboratory testing, and the results

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were processed using SoilSense. The application provided a clear interpretation of soil nutrient status and recommended the appropriate fertilizer requirements based on the measured nutrient levels. These recommendations helped confirm that the sowing medium possessed adequate nutrient availability to support early plant growth. Throughout the germination phase up to the fourth week of seedling development, the seedlings exhibited normal emergence, healthy cotyledon formation, and progressive leaf development, indicating that the soil conditions aligned with the app's assessment. These observations indicate that the soil conditions aligned with the fertilizer recommendations provided by SoilSense, demonstrating the system's usefulness in validating soil readiness before planting.

Transplantation and Early Vegetative Growth Stage



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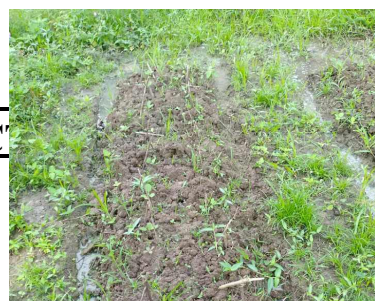
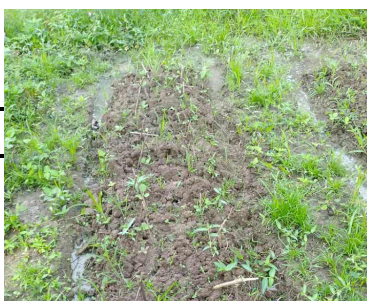
December 19, 2025 – Transplanting Stage (Day 0 DAT)

On December 19, 2025, the eggplant seedlings were transplanted from the seedling trays to the prepared soil plots. The three eggplant varieties—Casino, Long Eggplant, and Calixto—were planted in separate plots to ensure consistent observation and accurate comparison of growth performance. At the time of transplanting, the seedlings had well-developed stems and leaves, indicating readiness for field establishment.

The soil in all plots was freshly tilled and exhibited a dark, loose texture, suggesting adequate aeration and nutrient availability. Careful handling of the seedlings was observed during transplanting to minimize root disturbance and reduce the risk of transplant shock. This stage marked the transition from controlled nursery conditions to open-field cultivation, which is critical for subsequent vegetative growth.

Observation

During transplanting, all seedlings appeared healthy, upright, and free from visible damage. No immediate signs of stress such as leaf wilting or discoloration were observed. The uniform condition of the soil plots provided a suitable environment for root establishment across all three varieties.





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January 13, 2026 – Early Vegetative Growth (3 Weeks After DAT)

Three weeks after transplanting, noticeable changes in plant growth were observed across all experimental plots. The seedlings of Casino, Long Eggplant, and Calixto eggplants showed increased plant height, the emergence of new leaves, and improved stem strength. Differences in growth performance among the varieties became evident. Casino eggplant exhibited steady growth with moderate leaf expansion. Long Eggplant showed more vigorous vegetative growth characterized by broader leaves and faster height increase. Calixto eggplant displayed uniform growth with balanced leaf and stem development.

Discussion

The successful establishment of all three eggplant varieties after transplanting indicates that the soil preparation and transplanting procedures were effective. The absence of visible transplant shock suggests that the seedlings were transplanted at an appropriate developmental stage and that the soil conditions supported early root development.

More importantly, the healthy vegetative growth observed across Casino, Long Eggplant, and Calixto eggplants supports the reliability of the soil condition assessment provided by the SoilSense system. The measured soil parameters, including pH and essential nutrients,

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were within a suitable range for eggplant cultivation, as reflected by the plants' upright growth, continuous leaf production, and improved stem strength during the early vegetative stage.

The observed increase in plant height and leaf development three weeks after DAT signifies a successful transition from transplanting to active vegetative growth. Variations in growth performance among the three eggplant varieties may be attributed to genetic differences and their individual responses to soil nutrient availability. Despite these variations, all varieties demonstrated positive growth trends, indicating that the soil environment identified by SoilSense was generally favorable for eggplant growth.

Overall, these results demonstrate that the SoilSense system provided accurate and practical soil condition data that supported healthy plant establishment and early vegetative development. The consistency between SoilSense soil analysis and the actual growth response of the eggplant plants validates the system's effectiveness as a decision-support tool for crop cultivation. These findings further suggest that the plants are prepared for continued development in the succeeding stages, including flowering and fruit formation.

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Expected vs Actual Growth Timeline of Eggplant Based on SoilSense Fertilizer Recommendations

Growth Stage	Expected Timeline (Based on References & SoilSense Recommendations)	Actual Timeline (Observed in the Study)
Seed Germination	6–14 days after sowing	Germination observed within the expected period
Seedling Stage	4–8 weeks after germination	Seedlings developed healthy leaves and stems prior to transplanting
Transplanting Stage	30–45 days after sowing	Seedlings transplanted on December 19, 2025
Early Vegetative Growth	2–4 weeks after transplanting	Active vegetative growth observed 3 weeks after DAT (January 13, 2026)
Flowering Stage	50–60 days after sowing	<i>To be observed / ongoing</i>
Fruiting Stage	7–14 days after flowering	<i>To be observed / ongoing</i>
Harvest Maturity	100–120 days after sowing or 65–85 days after transplanting	<i>To be observed / ongoing</i>

Table 4.2. Expected vs Actual Growth Timeline of Eggplant

Effectiveness of SoilSense Fertilizer Recommendations

The consistency between the expected growth timeline and the actual plant development observed during the study confirms the effectiveness of the fertilizer recommendations provided by SoilSense. Healthy establishment, absence of nutrient deficiency symptoms, and continuous vegetative growth across all eggplant varieties indicate that the fertilizer



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guidance was appropriate and reliable. These results affirm SoilSense's capability as a fertilizer recommendation and decision-support system for sustainable crop management.

Soil Moisture Measurement

This study aimed to measure soil moisture and selected soil parameters in clay and loamy soils using a soil sensor. Measurements were taken before and after adding water to observe how each soil type responds to increasing moisture levels. By comparing changes in moisture content, temperature, pH, and nutrient levels, the experiment helps illustrate how soil texture influences water retention and nutrient behavior. The results can provide useful information for improving irrigation practices and soil management in agricultural systems.

Soil (Type)	Soil Parameters	Moisture Content	Temperature	ph	Nitrogen	Phosphorus	Potassium
Clay	Unit	%	oC	-	%	mg	meq/100g
	Sensor reading with inserted in the sample	43.34%	26.50 c	5.87	38.31 mg/kg	56.12 mg	101.20
	Sensor reading after adding a water	187.80%	25.90 c	6.32	48.40 mg/kg	26.95 mg	220.04
	Sensor reading after adding more water	197.37%	25.70 c	6.27	48.40 mg/kg	26.95 mg	220.04
Loamy	Sensor reading with inserted in the sample	39.31%	26.50 c	5.96	27.62 mg/kg	40.48 mg	73.00
	Sensor reading after adding a water	48.30%	25.70 c	6.37	23.10 mg/kg	13.04 mg	108.24
	Sensor reading after adding more water	48.30%	25.70 c	6.37	23.10 mg/kg	13.04 mg	108.24

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Discussion

The results show clear differences in moisture response and nutrient values between clay and loamy soils as water was added. In both soil types, moisture content increased after water addition, confirming that the sensor was able to detect changes in soil water status.

For clay soil, the initial moisture content was 43.34% when the sensor was inserted into the sample. After adding water, moisture increased sharply to 187.80% and further to 197.37% after adding more water. This large increase suggests that clay soil has a high water-holding capacity due to its fine particle size. Clay soil also showed higher nitrogen, phosphorus, and potassium values compared to loamy soil, indicating better nutrient retention. The pH values remained slightly acidic to near neutral, with only minor changes after watering.

In loamy soil, the initial moisture content was slightly lower at 39.31%. After adding water, moisture increased to 48.30% but did not increase further with additional water, suggesting faster drainage and lower water retention compared to clay soil. Nutrient values (nitrogen, phosphorus, and potassium) were lower than in clay soil, which is consistent with the looser structure of loam that allows nutrients to leach more easily. The pH values of loamy soil were slightly higher and more stable after watering.

Temperature remained relatively constant in both soil types, indicating that water addition had minimal effect on soil temperature during the experiment.

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Overall, the results demonstrate that clay soil retains more water and nutrients, while loamy soil shows quicker moisture stabilization after watering. This highlights the importance of soil type in moisture management and nutrient availability, especially for agricultural and irrigation practices.

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Variety Moisture Content

The researchers conducted this part of the study to determine how different varieties respond to changes in moisture content and related soil parameters. Moisture availability can influence temperature, pH, and nutrient levels, which are important factors affecting plant growth and development. Different varieties may vary in their ability to absorb and retain water, leading to differences in nutrient uptake.

Variety (Type)	Soil Parameters	Moisture Content	Temperature	ph	Nitrogen	Phosphorus	Potassium
Calixto	Unit	%	oC	-	%	mg	meq/100g
	Sensor reading with inserted in the sample	49.55%	24.10 c	6.37	34.10 mg/kg	19.13 mg	157.93
	Sensor reading after adding a water	59.55%	25.50 c	6.21	27.50 mg/kg	15.65 mg	127.76
	Sensor reading after adding more water	82.87%	25.50 c	5.97	27.50 mg/kg	15.65 mg	127.76
Casino	Sensor reading with inserted in the sample	68.71%	23.70 c	6.21	47.30 mg/kg	26.08 mg	212.94
	Sensor reading after adding a water	69.96%	25.40 c	6.27	22.00 mg/kg	12.17 mg	101.15
	Sensor reading after adding more water	94.52%	25.40 c	6.20	23.10 mg/kg	13.04 mg	106.47
Long Purple	Sensor reading with inserted in the sample	56.21%	23.40 c	6.18	33.00 mg/kg ¹	18.69 mg	152.61
	Sensor reading after adding a water	31.65%	25.30 c	6.40	7.70 mg/kg	4.35 mg	35.49
	Sensor reading after adding more water	74.12% ²	25.40 c	6.301	11.00 mg/kg	6.09 mg	51.46

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Discussion

The results show that moisture content and nutrient levels varied among the three varieties; Calixto, Casino, and Long Purple—when water was added. These differences indicate that each variety responds differently to changes in moisture availability.

For **Calixto**, the initial moisture content was 49.55% when the sensor was inserted into the sample. After adding water, moisture increased to 59.55% and further rose to 82.87% after additional water was added. This gradual increase suggests that Calixto has a good capacity to absorb and retain water. The nitrogen, phosphorus, and potassium values decreased after watering, which may be due to dilution effects or nutrient movement with water. The pH slightly decreased, indicating a mild increase in acidity as moisture increased.

In **Casino**, moisture content was already high at 68.71% during the initial reading. Only a small increase was observed after the first water addition (69.96%), but a larger increase occurred after adding more water (94.52%). This suggests that Casino may require higher water input before showing a strong moisture response. Nutrient levels, particularly nitrogen and potassium, showed noticeable changes after watering, indicating sensitivity to water availability.

For **Long Purple**, the initial moisture content was 56.21%, but a decrease was observed after the first water addition (31.65%). This unusual drop may be due to uneven water distribution, sensor placement error, or sample disturbance during measurement. After adding more water, moisture increased to 74.12%, showing that Long Purple can still

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absorb water when sufficient moisture is supplied. Nutrient values in this variety were generally lower than in Calixto and Casino, suggesting lower nutrient retention.

Across all varieties, temperature remained relatively stable, showing minimal influence from water addition. Overall, the findings indicate that moisture response and nutrient behavior differ among varieties, emphasizing the importance of considering varietal differences in moisture management and agricultural practices.

CHAPTER 5

SUMMARY, FINDINGS, CONCLUSIONS, AND RECOMMENDATIONS

This chapter presents a comprehensive summary of the findings, conclusions, and recommendations derived from the development, testing, and evaluation of *SoilSense*, an Internet of Things (IoT) and machine learning–based soil health classification and fertilizer recommendation system. The results obtained from Chapter 4, which include the demographic characteristics of the respondents, system functionality, sensor performance, and the system’s

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quality assessment based on the ISO/IEC 25010 software quality model, serve as the basis for the analyses and insights discussed herein.

5.1 Summary of Findings

The study focused on the design, development, and evaluation of the SoilSense system. Also, the key findings of this study conducted among 36 farmers of RIC Organization in Brgy Rizal Rural Lopez, Quezon regarding the demographic profile of users and the system quality of SoilSense ISO/IEC 25010 standards. By examining both demographic data and user assessment, this summary provides a clear understanding how SoilSense performs in a real-world conditions and highlights areas that requires further improvement to better support agricultural practices.

5.1.1 Demographic Profile of Respondents

The evaluation involved **36 farmer-respondents**, with a majority identifying as male (55.56%), followed by female participants (41.67%). Most respondents belonged to the **50–59 age group (27.78%)**, while the fewest were in the 20–29 and 70–79 categories (2.78%). In terms of agricultural experience, a substantial portion had **6–10 years** or **over 21 years** of farming experience. The predominant eggplant variety cultivated was **Calixto**, and all respondents reported **non-utilization of irrigation systems**, indicating reliance on manual or rainfall dependent watering practices.

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5.1.2 ISO/IEC 25010 System Quality Evaluation

The overall weighted mean across all ISO 25010 criteria was **4.17 (Agree)**, signifying that respondents assessed SoilSense as generally effective and satisfactory in fulfilling its intended functions.

1. Functionality (4.33 – Strongly Agree)

The system was rated highly in accurately measuring soil pH, NPK, moisture, temperature, and humidity. Minor concerns were raised regarding the alignment of crop and fertilizer recommendations with localized field conditions.

2. Performance Efficiency (4.10 – Agree)

Respondents noted acceptable processing speed and responsiveness. However, occasional delays were observed in near-real-time sensor updates and mobile application loading, particularly on lower-end devices.

3. Compatibility (4.06 – Agree)

SoilSense was perceived as compatible with existing farming practices. Nonetheless, some participants encountered issues related to the system's compatibility with varying Android device models and software versions.

4. Usability (4.12 – Agree)

The system was considered easy to use and beneficial for supporting farming decisions. Lower scores were recorded for interface clarity and ease of navigation, indicating the need for interface refinement.

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5. Reliability (3.94 – Agree)

Sensor readings were generally stable. However, reduced performance was noted during intermittent network connectivity, suggesting the need for improved system resilience.

6. Security (4.35 – Strongly Agree)

Participants expressed strong confidence in the system's capacity to safeguard personal and farm-related data, facilitated in part by the absence of login requirements.

7. Maintainability (4.20 – Agree)

Users found system updates manageable. Minor challenges remained in calibration and device maintenance procedures.

8. Portability (4.24 – Strongly Agree)

The system could be reinstalled or relocated with relative ease. A few respondents experienced difficulties when repositioning sensors across different fields.

5.2 Conclusions

Based on the evaluation outcomes, the following conclusions were drawn:

The SoilSense system effectively achieves its primary objectives of real-time soil health assessment and fertilizer recommendation generation. The high functionality and security ratings demonstrate its capacity to operate accurately and securely within agricultural settings.

The system substantially contributes to informed agricultural decision-making, as evidenced by positive usability assessments. The majority of respondents indicated that the system enhanced their understanding of soil conditions and supported fertilizer management practices.

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Areas requiring enhancement remain, particularly regarding real-time sensor responsiveness, device compatibility, and system behavior under unstable internet connectivity. Addressing these limitations will significantly improve operational efficiency and user satisfaction.

Data privacy and accessibility are notable strengths of the system. Respondents' strong confidence in the system's data protection mechanisms underscores its suitability for farmers who may be cautious about digital platforms.

Given its ease of use, portability, and reliable measurement capabilities, SoilSense is a viable technological intervention for small-scale farmers, even in communities with limited digital literacy or technological infrastructure.

5.3 Recommendations

5.3.1 Recommendations for System Improvement

1. Enhance real-time data transmission and processing

Optimize communication protocols between sensor nodes and the mobile application.

Implement improved data buffering mechanisms to sustain performance during connectivity disruptions.

2. Strengthen device and operating system compatibility

Conduct comprehensive testing across a broader range of low-end and mid-tier Android devices. Implement adaptive screen scaling and UI responsiveness to accommodate varying screen sizes.

3. Refine the user interface design

Adopt a more structured layout, intuitive icons, and standardized visual elements.

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Incorporate interactive onboarding modules or in-app guidance for new users.

4. Improve system reliability and offline operation

Develop offline-capable modules enabling the application to store and synchronize data once connectivity is restored. Provide clearer calibration instructions and routine maintenance prompts.

5. Localize fertilizer recommendations

Integrate crop and soil guidelines from sources such as the Department of Agriculture or local agricultural extension services. Allow users to specify region-based variables or crop varieties to enhance recommendation accuracy.

5.3.2 Recommendations for System Implementation

1. Deployment in agricultural communities

Introduce SoilSense through farmer organizations, cooperatives, and local government agricultural programs to increase adoption.

2. Capacity-building activities

Conduct field demonstrations and instructional workshops focusing on device operation, sensor maintenance, calibration, and data interpretation.

3. Collaboration with agricultural agencies

Integrate SoilSense into extension programs to support soil profiling, monitoring, and crop advisory services.

5.3.3 Recommendations for Future Researcher

1. Integration of automated irrigation systems

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Expand the system to include IoT-based irrigation control guided by soil moisture thresholds.

2. Advancement of machine learning models

Enhance the existing model to support additional crops, detect soil-borne diseases, and generate predictive yield estimations.

3. Support for additional soil types and field conditions

Broaden the dataset through multi-regional soil sampling and long-term monitoring.

4. Longitudinal field studies

Conduct extended evaluations to compare the productivity of farms utilizing SoilSense against those employing traditional practices.

5. Development of a centralized web-based dashboard

Implement a cloud-enabled platform for long-term data visualization, trend analysis, and decision-support analytics.

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APPENDICES

APPENDIX A

DATA GATHERING INSTRUMENT

Capstone Title: Soilsense: Machine Learning-Based Soil Health Classification and Crop Fertilizer Recommendation System

Dear Respondent,

The researchers developed SoilSense: Soilsense: Machine Learning-Based Soil Health Classification and Crop Fertilizer Recommendation System and this will be evaluated by target respondents using this survey instrument. The rating you will provide will help the researchers conclude the acceptability of the proposed software in attaining the objectives of this study.

Thank you very much for your participation.

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Part I: Demographic Profile
(Optional – for research classification only)

Name (Optional): _____

Sex: ☐ Male ☐ Female ☐ Prefer not to say

Age: _____

Years of Experience as Farmer:

☐ Less than 5 year ☐ 6–10 years ☐ 11–15 years
☐ 16–20 years ☐ More than 21 years ☐ Others: _____

Eggplant Variety Type: _____

Agricultural Practices: ☐ Irrigation ☐ Not Irrigation

Instructions: Rate the following based on your experience with the IoT-based system.

- (5) Excellent / Strongly Agree
- (4) Good / Agree
- (3) Neutral
- (2) Poor / Disagree
- (1) Very Poor / Strongly Disagree



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Part II: System Quality Evaluation (ISO 25010)

SYSTEM ACCEPTABILITY CHARACTERISTICS		RATING				
Functionality (Does the system do what it's supposed to do?)		5	4	3	2	1
1. The system accurately measures soil pH, nitrogen, phosphorus, potassium, temperature, and humidity.						
2. The crop and fertilizer recommendations are relevant to local conditions.						
3. All features described in the user guide are present and work as intended.						
Performance Efficiency (How fast and responsive is the system?)						
The system processes data quickly and provides timely recommendations.						
2. Sensor readings update in near real time without noticeable delay.						
3. The mobile app loads quickly and uses minimal mobile data.						
Compatibility (Does it work well with my devices and setup?)						
1. The app works well on my smartphone model and Android version.						
2. <u>SoilSense</u> integrates smoothly with my existing farming practices.						
Usability (How easy is it to use?)						
1. The interface is clear and easy to navigate.						
2. Icons, buttons, and instructions are easy to understand.						
3. I can learn to use the system without outside help.						
4. <u>SoilSense</u> improves my farming decisions.						
5. I am satisfied overall with the system.						
Reliability (Does it work consistently?)						
1. Sensor data remains consistent during repeated measurements.						
2. The system continues to function even after short internet outages.						
Security (Is my data safe and protected?)						
1. I am confident my personal and farm data are protected and private.						
2. Having no login process prevents providing sensitive and personal information.						
Maintainability (Can it be easily maintained and updated?)						
1. System updates or fixes can be installed easily.						
2. Device maintenance and calibration are simple.						
Portability (Can it be easily installed and adapted?)						
1. I can move the sensor to another field and reconnect it easily.						
2. The app can be reinstalled on another phone without issues.						

Part III: Suggestions for Improvement

What improvements would you suggest for the SoilSense system?

APPENDIX B

EVALUATION TOOL, TEST DOCUMENTS, AND TEST RESULTS

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 REPUBLIC OF THE PHILIPPINES
POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
OFFICE OF THE VICE PRESIDENT FOR CAMPUSES
LOPEZ QUEZON CAMPUS



October 26, 2025

Hon. ILLAN A. NAVAL
Begy, Captain Rizal-Rural

THRU: RIZABEL ROMERO TARACINA
RIC, President

Subject: Request for Deployment of Capstone Project

Dear Hon. Naval,

We, the fourth year Bachelor of Science in Information Technology students (BSIT 4-1) from the Polytechnic University of the Philippines, are writing to seek your permission to deploy our Capstone Project entitled, "SoilSense: Machine Learning-Based Soil Health Classification and Crop Fertilizer Recommendation System" within premises of the Begy, Rizal-Rural, Lopez, Quezon.

This project aims to support sustainable agriculture by leveraging machine learning and Internet of Things (IoT) technologies to assess soil health based on measurable parameters such as pH, nitrogen (N), phosphorus (P), potassium (K), and moisture levels. Through this system, users can receive real-time data and recommendations for optimal fertilizer usage.

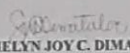
The objectives of our capstone project include:

- To develop a mobile app that can collect and preprocess soil datasets with key features (pH, nitrogen, phosphorus, potassium, humidity).
- To design a user-friendly Android application for farmers and agricultural stakeholders.
- To implement a rule-based or ML-aided recommendation engine for appropriate fertilizer and crop suggestions based on soil properties and classification.

We respectfully seek your approval to deploy and test the system on your premises for research and validation purposes. We are ready to provide a detailed project overview and address any concerns you may have regarding the implementation. Your support will be instrumental in helping us assess the effectiveness and potential impact of this system in an actual agricultural environment.

Thank you for considering our request. We look forward to your positive response.

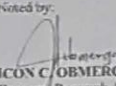
Sincerely,


WENIELYN JOY C. DIMATULAC
Researcher


PRINCESS JANE R. JIMENEZ
Researcher


JODELLE N. LUNA
Researcher


LAIZEL PIA M. NONZOL
Researcher

Noted by:

ICOX C. OBERMGA
Capstone Research Adviser

2nd Floor Administrative Bldg., Yumal St. Brgy., Burgos, Lopez, Quezon
Website: www.pup.edu.ph | Inquiries: help@pup.edu.ph

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The researchers present the SoilSense system to the farmers of RIC with the Brgy. Official.

After that session the researchers visit again to conduct a survey.



ON





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In Barangay Rizal, Lopez, Quezon, the researchers established three experimental plots, planting Calixto, Casino, and Long Purple varieties in each plot.

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APPENDIX C

USER MANUAL

SoilSense User Manual

1. Introduction

SoilSense is a soil monitoring and fertilizer recommendation system that combines IoT sensors and machine learning to help farmers make data-driven decisions. It measures key soil properties such as pH, nitrogen, phosphorus, potassium, temperature, and humidity, and provides real-time crop and fertilizer recommendations directly to a mobile app.

2. System Components

The SoilSense system consists of hardware and software components:

- Soil Comprehensive Sensor – A multi-parameter sensor that measures soil pH, nitrogen, phosphorus, potassium, temperature, and humidity.
- ESP32 Microcontroller – Serves as the central controller, reading sensor data and transmitting it to the cloud.
- Power Supply – Provides stable power to the sensor and microcontroller.
- Mobile Application – An Android app that displays real-time soil readings and provides recommendations.
- Firebase Cloud – Stores and manages data securely and supports real-time updates.

3. Setup Guide

Follow these steps to set up the SoilSense system:

1. Unbox and ensure all components are included: Soil Comprehensive Sensor, ESP32, power supply, and necessary cables.
2. Connect the sensor to the ESP32 microcontroller using the provided wiring diagram.
3. Power on the ESP32 and ensure the sensor is properly connected.
4. Connect the ESP32 to a stable Wi-Fi network.
5. Download and install the SoilSense mobile application on your Android device.
6. Pair the mobile application with the ESP32 by following the on-screen prompts.
7. Calibrate the sensor if necessary, using the calibration feature in the mobile app.

4. Using the System

Once set up, the SoilSense system operates automatically to monitor soil health:

- Open the mobile application to view real-time soil data.
- Review current readings for pH, nitrogen, phosphorus, potassium, temperature, and humidity.
- Receive automated crop and fertilizer recommendations based on the analyzed data.
- Follow the fertilizer application and crop selection guidance provided to improve soil and crop productivity.

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5. Maintenance and Troubleshooting

To maintain optimal performance:

- Clean the sensor probe regularly to prevent soil residue build-up.
- Check the power supply and Wi-Fi connection periodically.
- Recalibrate the sensor every month or after extreme weather conditions.

Common issues:

- No Data Displayed: Ensure the ESP32 is powered and connected to Wi-Fi.
- Inaccurate Readings: Recalibrate the sensor and verify proper installation.
- App Errors: Restart the app or reinstall if necessary.

6. Data Privacy and Security

The SoilSense system uses end-to-end encryption to protect all transmitted and stored data. User accounts are protected and access to soil data is restricted to authorized users only. Data is stored securely in Firebase and automatically backed up.

7. Updates

The SoilSense team periodically releases updates to enhance features and security:

- Ensure your mobile app is set to auto-update or check the app store regularly.
- Firmware updates for the ESP32 can be installed using the mobile app's update function.
- Back up data before major updates to prevent data loss.



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APPENDIX D

CURRICULUM VITAE

Dimatulac, Wenielyn Joy C.

Brgy. Talolong Lopez, Quezon

Email: wenielynjoydimatulac03@gmail.com

Contact Number: 09669421018



Personal Information

Date of Birth: June 03, 2003

Age: 22

Place of Birth: Lopez, Quezon

Citizenship: Filipino

Mother: Elenita C. Dimatulac

Father: Pedro M. Dimatulac

Religion: Roman Catholic

Civil Status: Single

Gender: Female

Education

Tertiary: Polytechnic University of the Philippines 2022 - Present
Yumul St. Brgy. Burgos Lopez Quezon
Bachelor of Science in Information Technology

Secondary: Magallanes National High School 2018-2020
Magallanes Lopez, Quezon
TVL – CSS (Computer System Services)

Junior: Magallanes National High School
2014-2018
Magallanes Lopez, Quezon

Primary: Villa Aurora Elementary School 2008-2014
Brgy. Villa Aurora Lopez, Quezon

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Jimenez, Princess Jane R.

Brgy. Bocboc Lopez, Quezon

Email: princessjanejimenez917@gmail.com

Contact Number: 0956-517-3473



Personal Information

Date of Birth: July 19, 2002

Age: 23

Place of Birth: Quezon, Quezon

Citizenship: Filipino

Mother: Imelda R. Jimenez

Father: Pedro V. Jimenez

Religion: Iglesia Ni Cristo

Civil Status: Single

Gender: Female

Education

Tertiary: Polytechnic University of the Philippines 2022 - Present
Yumul St. Brgy. Burgos Lopez Quezon
Bachelor of Science in Information Technology

Secondary: Cesar Tam Memorial National High School
2019-2021
Barangay 4 Quezon, Quezon
TVL - Home Economics

Junior: Cesar Tam Memorial National High School
2015-2019
Barangay 4 Quezon, Quezon

Primary: Quezon Elementary School 2009-2015
Barangay 4 Quezon, Quezon

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Luna, Jodell N.

Brgy. Binutas Calauag, Quezon
Email: jodell.luna14@gmail.com
Contact Number: 0945-504-2349



Personal Information

Date of Birth: June 04, 2004
Age: 21
Place of Birth: Magarao, Camarines Sur
Citizenship: Filipino
Mother: Eudelia B. Nantes
Father: Emmanuel O. Luna
Religion: Roman Catholic
Civil Status: Single
Gender: Male

Education

Tertiary: Polytechnic University of the Philippines 2022 - Present
Yumul St. Brgy. Burgos Lopez Quezon
Bachelor of Science in Information Technology

Secondary: Calauag Central College Inc. 2021-2022
JP. Rizal St., Arguelles Cor., Brgy. Poblacion Cuatro, Calauag Quezon
Humanities and Social Sciences

Junior: Calauag Central College Inc.
2019-2020
JP. Rizal St., Arguelles Cor., Brgy. Poblacion Cuatro, Calauag Quezon

Primary: Binutas Elementary School 2015-2016
Brgy. Binutas Calauag, Quezon

CAPSTONE PROJECT



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES

Nonzol, Laizel Pia M.

Brgy. Tabugon Sta. Elena Camarines Norte

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Personal Information

Age: 21
Place of Birth: Tanay, Rizal
Citizenship: Filipino
Mother: Melinda G.C Mag-isa
Father: Jose Elizaldy V. Nonzol
Religion: Roman Catholic
Civil Status: Single
Gender: Female

Education

Tertiary: Polytechnic University of the Philippines 2022 - Present
Yumul St. Brgy. Burgos Lopez Quezon
Bachelor of Science in Information Technology

Secondary: Lyceum of St. Dominic Inc. 2020-2022
Sta. Elena Camarines Norte
TVL – CSS (Computer System Services)

Junior: Leocadio Alejo Entienza Highschool
2016-220
Sitio Lobo Brgy. Tabugon Sta. Elena Camarines Norte

Primary: Tabugon Elementary School
2018-2016
Brgy. Tabugon Sta. Elena Camarines Norte

